Computer Architecture
Informatics Eng., ESTiG/IPB
2024/2025

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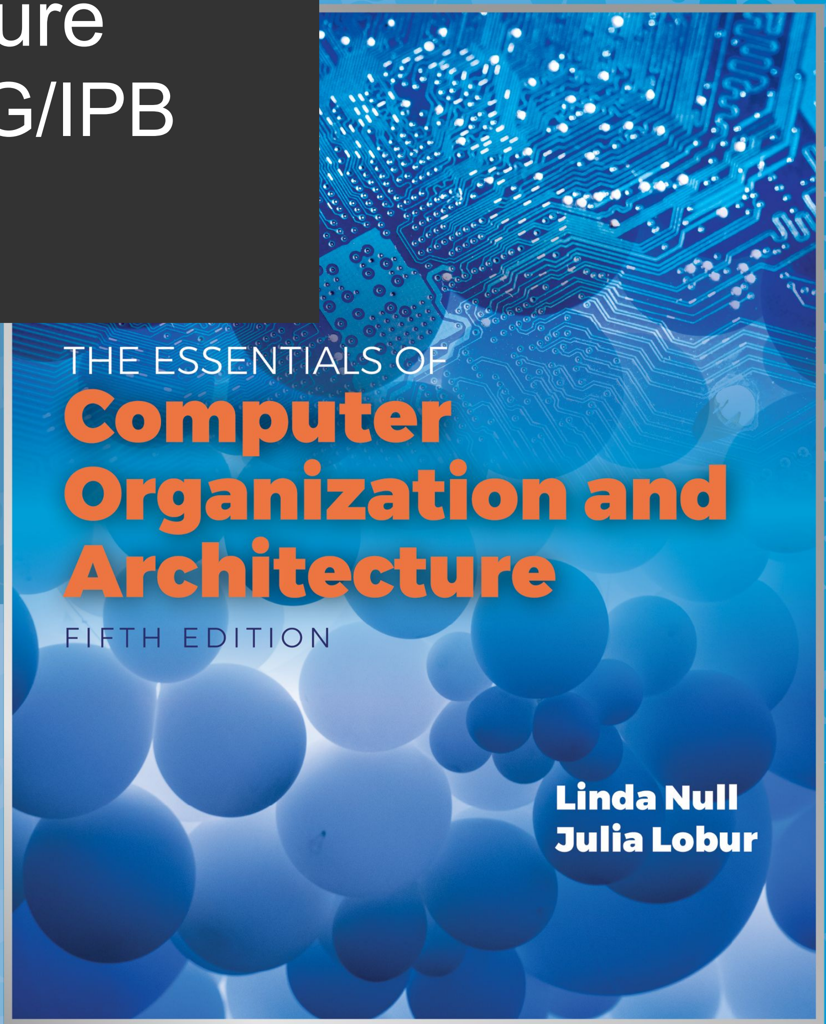
(mainly based on Chapter 7 of the book "The Essentials of Computer Organization and Architecture" of Linda Null & Julia Lobur)

Unit 6:
Input/Output and Storage Systems



Objectives

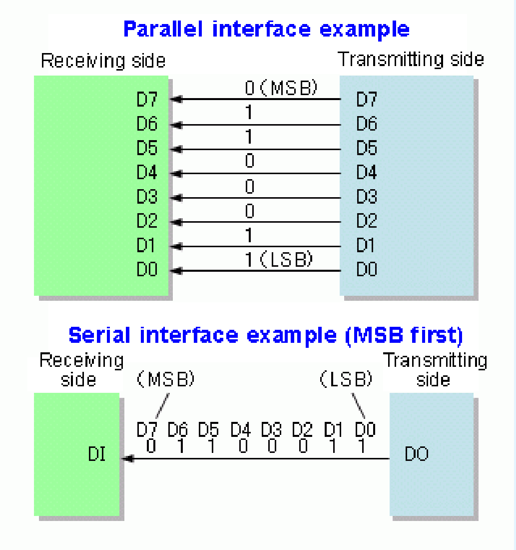
- Know the kinds of buses and their operation principles
- Understand the concept and goal of CPU interrupts
- Understand how I/O systems work, including I/O methods and architectures.
- Become familiar with storage media, and the differences in their respective formats.
- Understand the essential ideas underlying data compression, and become familiar with different types of compression algorithms.

6.1 Introduction

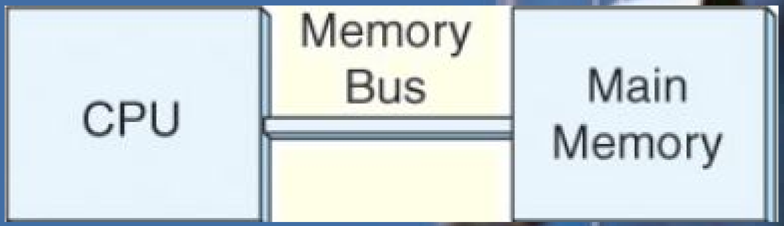
- Any computer has multiple buses; it is important to know their main types and operation basic principles
- All computers have also I/O devices connected to them, used to interact with the “outside world”
- In studying I/O, we seek to understand the different types of I/O devices as well as how they work.
- Data storage and retrieval is one of the primary functions of computer systems, therefore it is important to know the essentials of these functions.

6.2 Buses

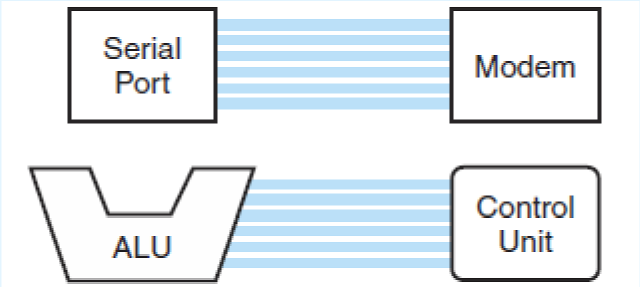
- **Bus**:
  - set of "conducting lines" that carry (encoded) between components of a computing / communication system
  - bus speed depends on the *width* and *number of devices connected*
  - several configurations
    - **parallel bus**: several bits at the same time
    - **serial bus**: one bit a time (in sequence)
    - **point-to-point / dedicated bus**: links two componentes/devices
    - **multi-point (shared)**: links several devices (only one may write into the bus)
    - ***primary / secondary*** devices: initiate actions / react to actions



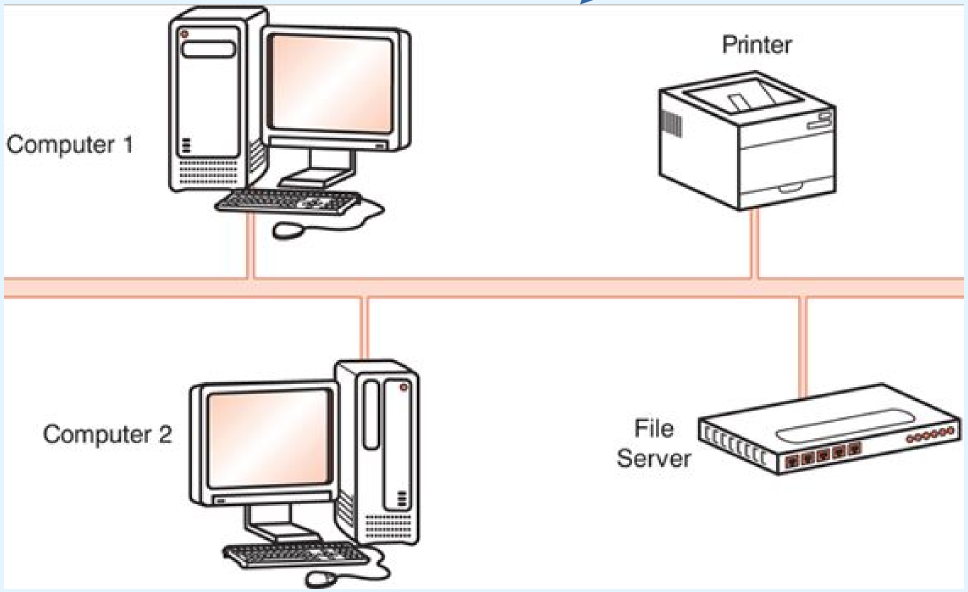
6.2 Buses

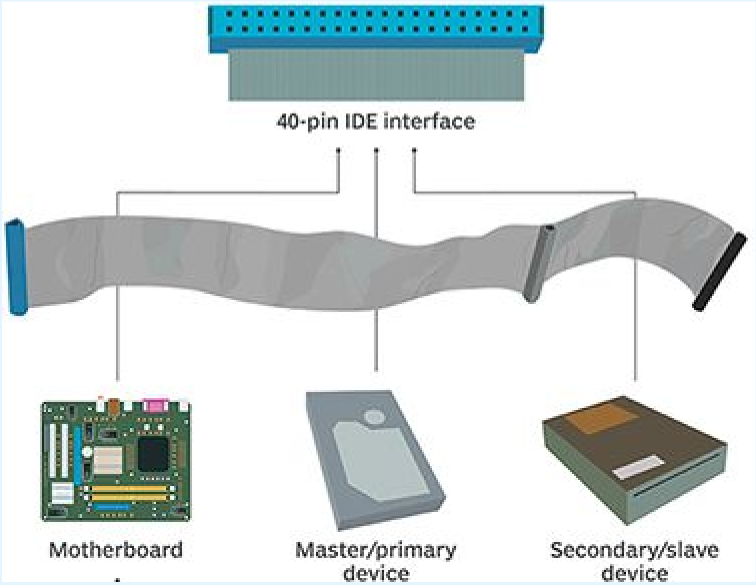


» point-to-point buses:

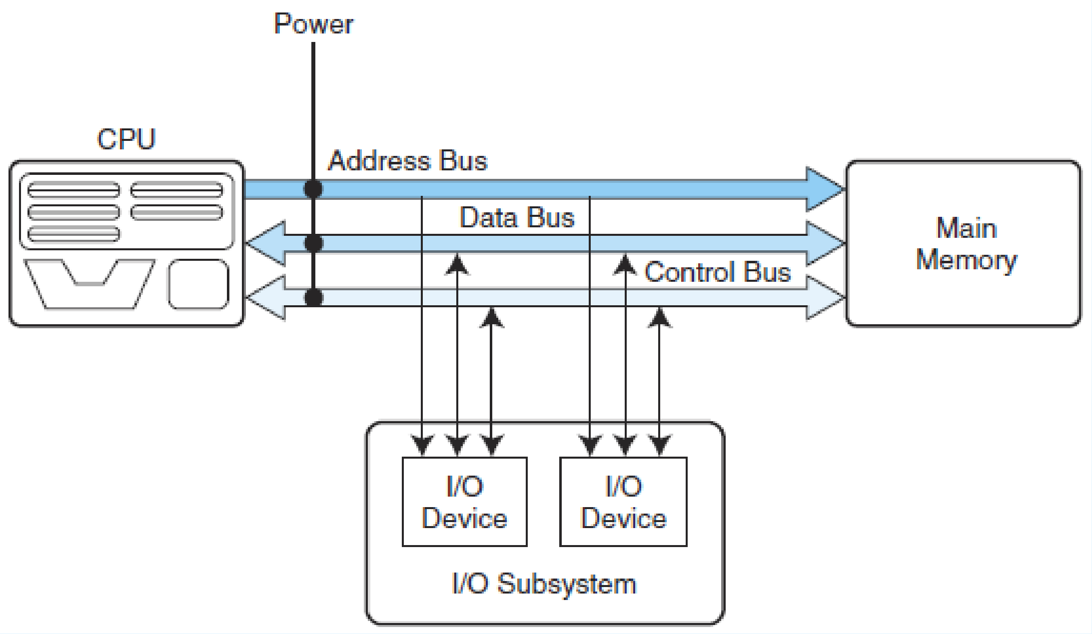


» multi-point buses:





6.2 Buses



- ***data lines***:

  - carry *data bits* between devices

- ***control lines***:

  - set the direction of data flow, and when each device can access the bus

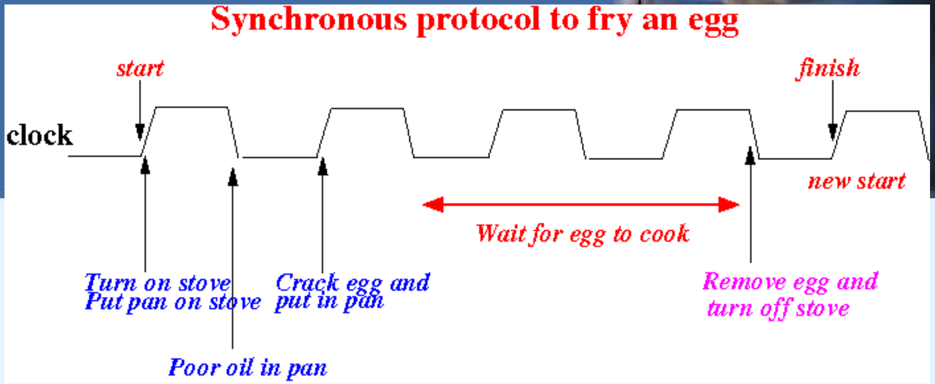
- ***address lines***:

  - set the *location* of the source / destination of the data

6.2 Buses

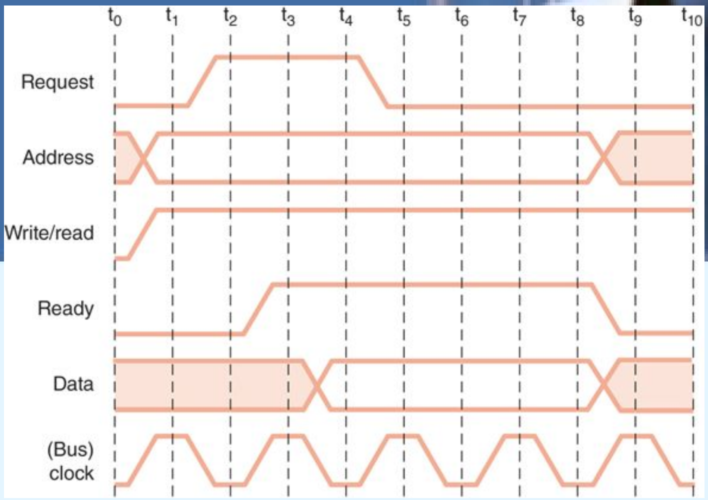
- ***CPU – Main Memory bus:***
  - short length, (very) high speed, specific usage
- ***Input/Output (I/O) buses:***
  - wider, slower, generic usage (different devices with different transfer speeds may share the same I/O bus)
  - USB, Thunderbolt, IDE, SCSI, SATA, SAS, PCI, PCIe
- ***backplane buses:***
  - *storage*: *backplane* used for direct connection (no cables) of multiple hot-swapable disks (Hard Disks or SSDs)
  - *blade servers*: *backplane* in the back of a server case used to connect multiple thin-motherboards (blades)

6.2 Buses



- ***synchronous buses:***
  - the bus operation is guided just by the internal **bus clock**
  - operations initiated in the transition between clock cycles
  - all devices connected to the bus synchronized to bus clock
  - *clock skew/drift* may lead to problems (wrong timing of operations) - bit-cell timing limits are not rigorous
  - with shorter buses, these skews will be smaller
  - transmission time must be <= bus clock cycle time
  - thus, the bus width limits the bus clock frequency

6.2 Buses



- ***assynchronous buses:***
  - operations controlled through control lines and a **handshaking protocol**
  - example (reading data from a memory)
    - the control line **ReqRead** is activated and the address bits are placed in the address lines
    - the control line **ReadyData** is activated when the data read is placed by the memory circuitry in the data lines
    - the control line **ACK** is activated to signal that the operations ReqRead e ReadyData where confirmed
  - using a protocol in addition to a clock allows for higher performance and support for a wider range of devices

6.2 Buses

- In a *primary-secondary* configuration, where more than one device can be primary (the bus master), concurrent master requests must be arbitrated.

- Four **categories of bus arbitration** are:
  - **Daisy chain:** Permissions are passed from the highest-priority device to the lowest.
  - **Centralized parallel:** Each device is directly connected to an arbitration circuit.
  - **Distributed using self-detection:** Devices decide which gets the bus among themselves.
  - **Distributed using collision-detection:** Any device can try to use the bus. Tries again if its data collides with another device's data.

6.3 Interrupts

- The normal execution of a program is altered when an event of higher-priority occurs. The CPU is alerted to such an event through an **interrupt** mechanism.
- Interrupts can be triggered by I/O requests, arithmetic errors (such as division by zero), or when an invalid instruction is encountered (wrong/unauthorized opcode).
- **Interrupt Servicing**: the Operating System saves the current state (register content) of the CPU and executes the **interrupt service routine (ISR)** specified in the interrupt vector; once the ISR is concluded, the CPU restores the saved state and continues the program.

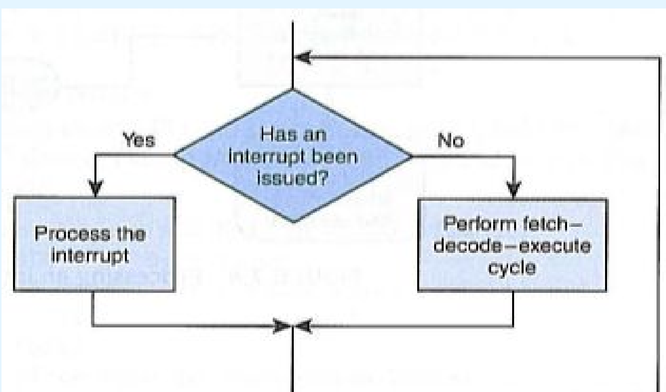
6.3 Interrupts

Main Memory (0 .. M-1)

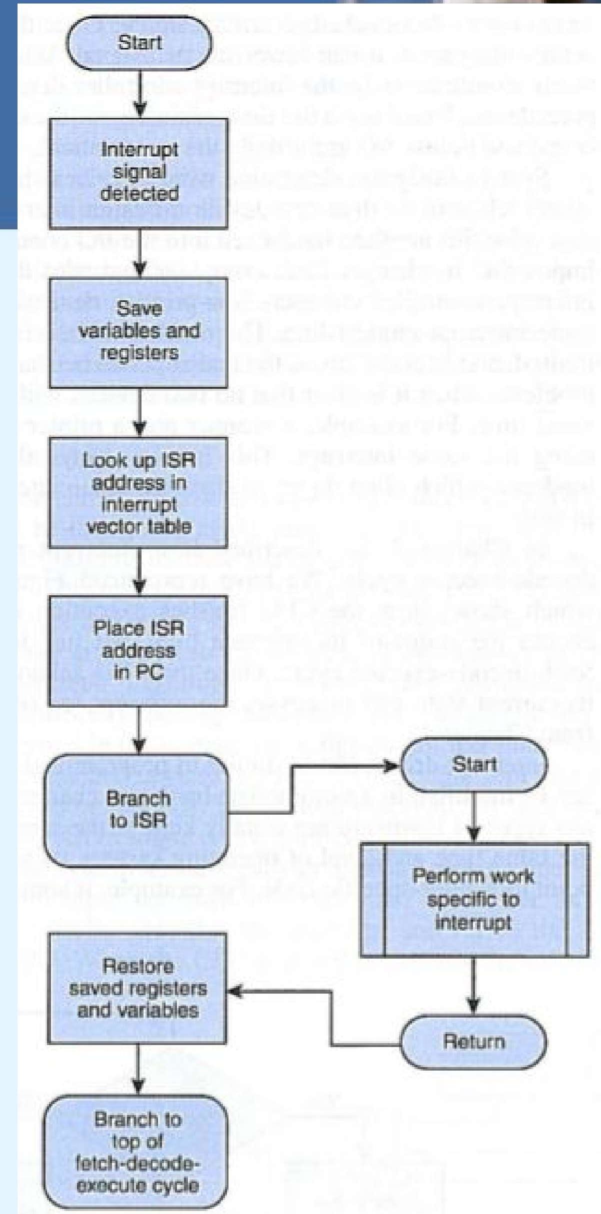
Address	Content
0	1000
1	2000
…	…
255	10000
…	…
1000 …	ISR0
2000 …	ISR1
	…
M-1	…

IV

Interrupt Vector (IV) with support to 256 interrupts.



fetch-decode-execute cycle that supports interrupts



Interrupt Processing

6.4 I/O Architectures
- I/O Subsystems

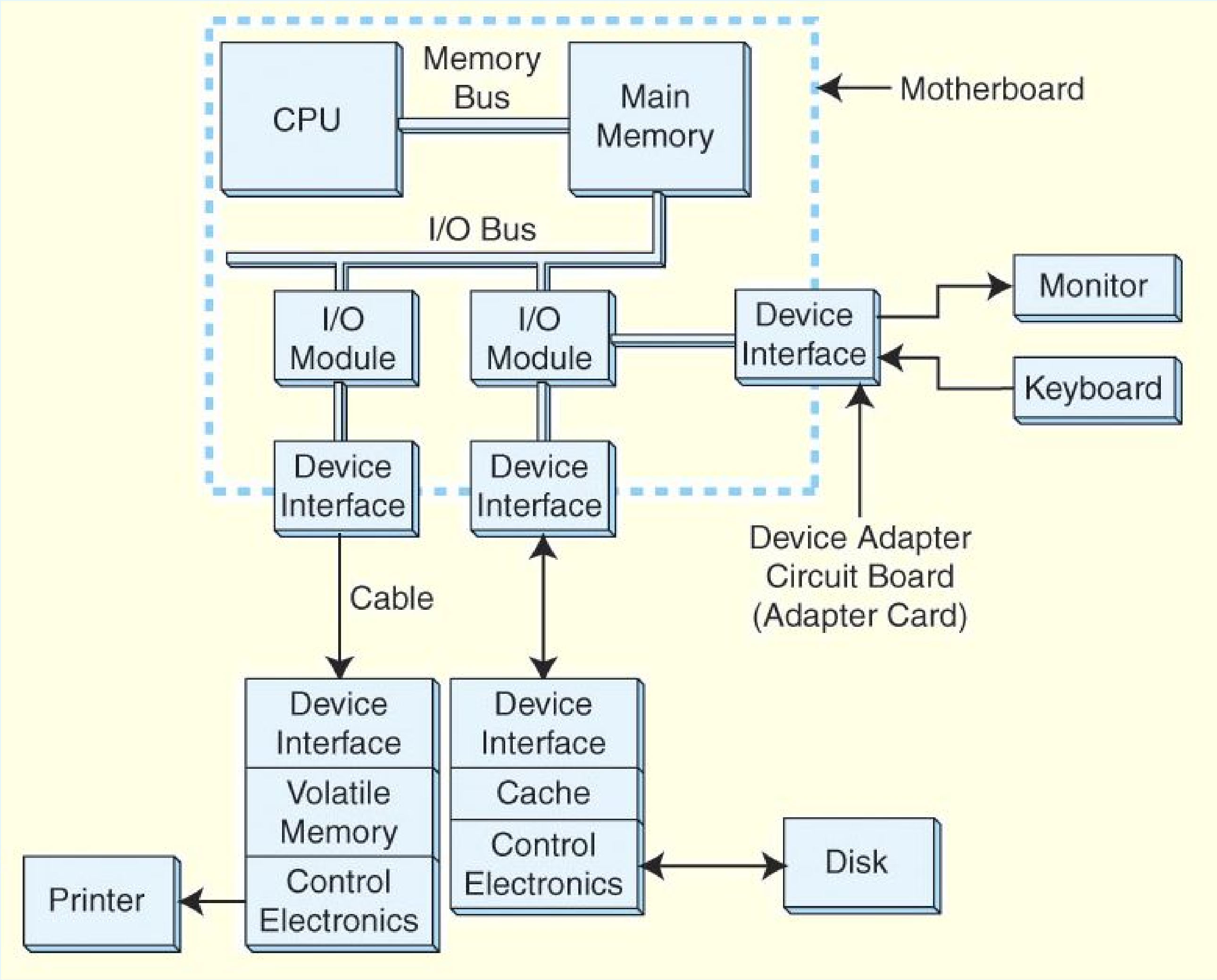
- A computer communicates with the outside world through its input/output (I/O) subsystem.
- I/O devices (peripherals) are connect to the CPU through various interfaces (USB, SATA, PCI, ...
- I/O can be based on
  – polling (instruction-based, where the CPU has a specialized I/O instruction set, or memory-mapped, where the I/O device behaves like main memory from the CPU point of view)
  – interrupts, DMA, IO processors

6.4 I/O Architectures
- I/O Subsystems

- **Input/Output subsystem**: set of components that move coded data between external devices and a host system (host system = CPU + main memory)
- I/O subsystems include components like:
  – Blocks of main memory that are devoted to I/O functions.
  – Buses that move data into and out of the system.
  – Control modules in the host and in peripheral devices,
  – Interfaces to external components such as keyboards and disks.
  – Cabling or communications links between the host and peripherals.

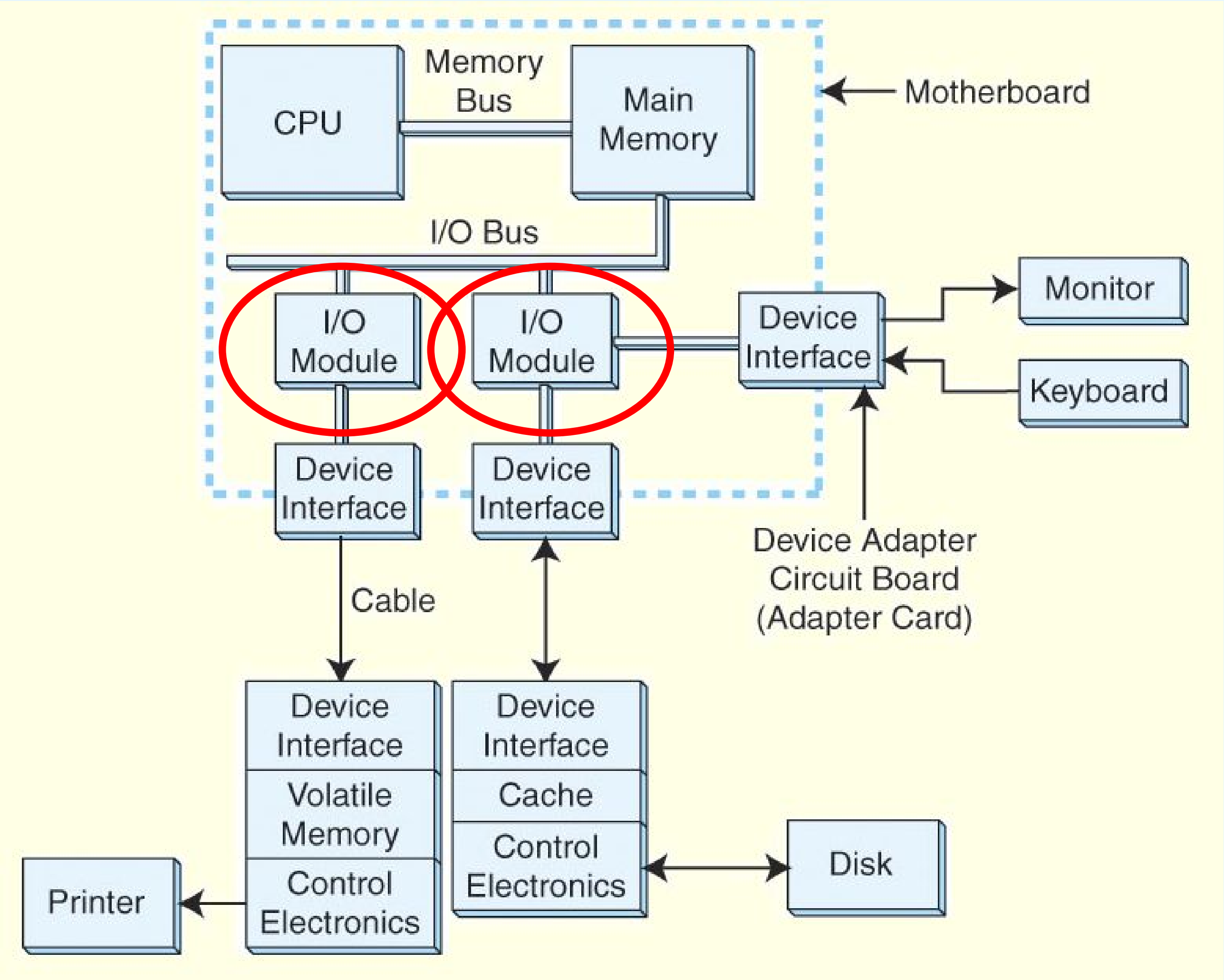
6.4 I/O Architectures
- I/O Subsystems

- A model IO configuration



6.4 I/O Architectures
- I/O Subsystems

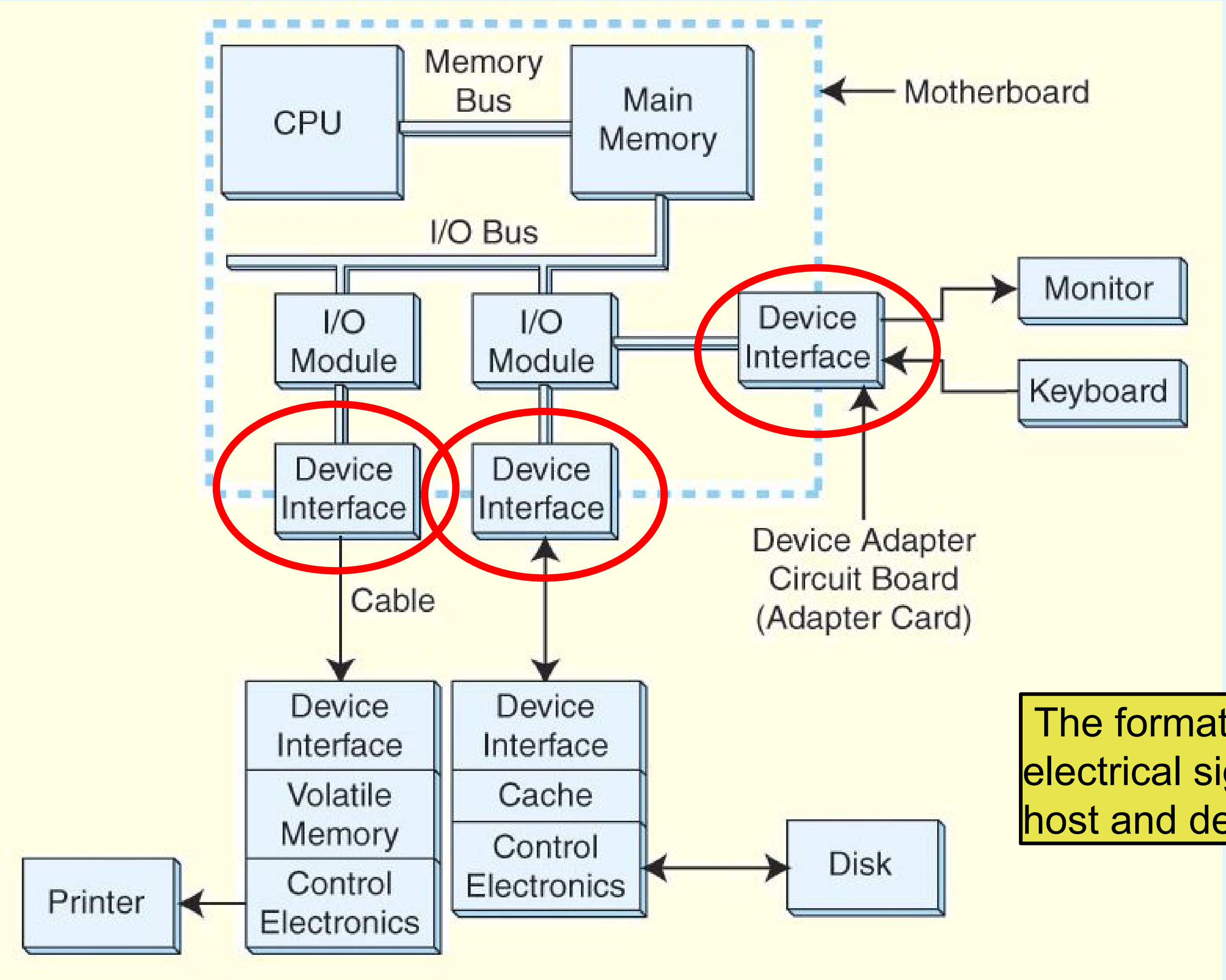
- A model IO configuration (cont.)



  – **I/O modules** move data between the **main memory** and specific **device interfaces** (disk, printer, ...)

6.4 I/O Architectures
- I/O Subsystems

- A model IO configuration (cont.)

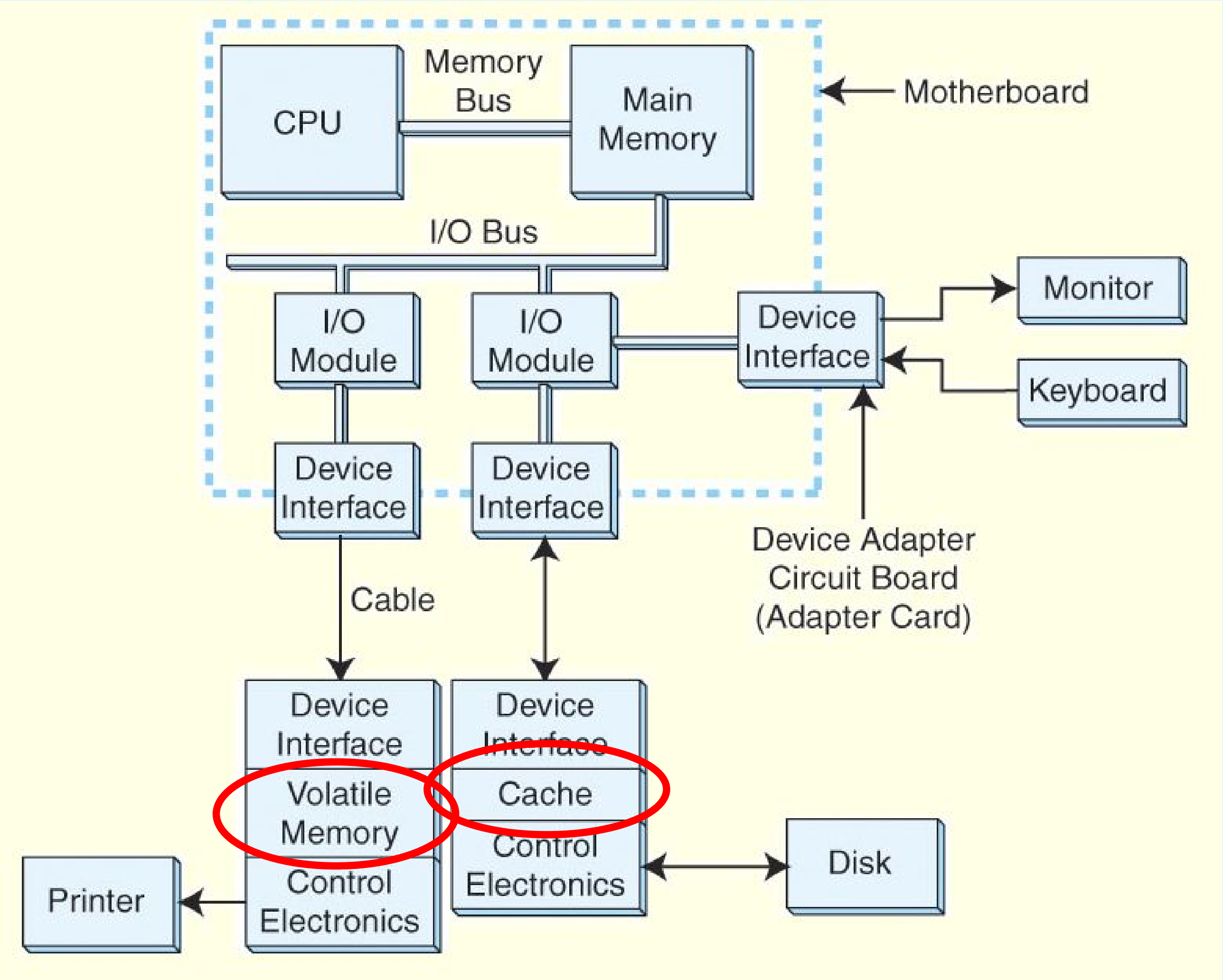


  - the **device interfaces** make sure that the host and the device are ready to communicate

The format and the meaning of the electrical signals exchanged between host and devices obeys to a **protocol**

6.4 I/O Architectures
- I/O Subsystems

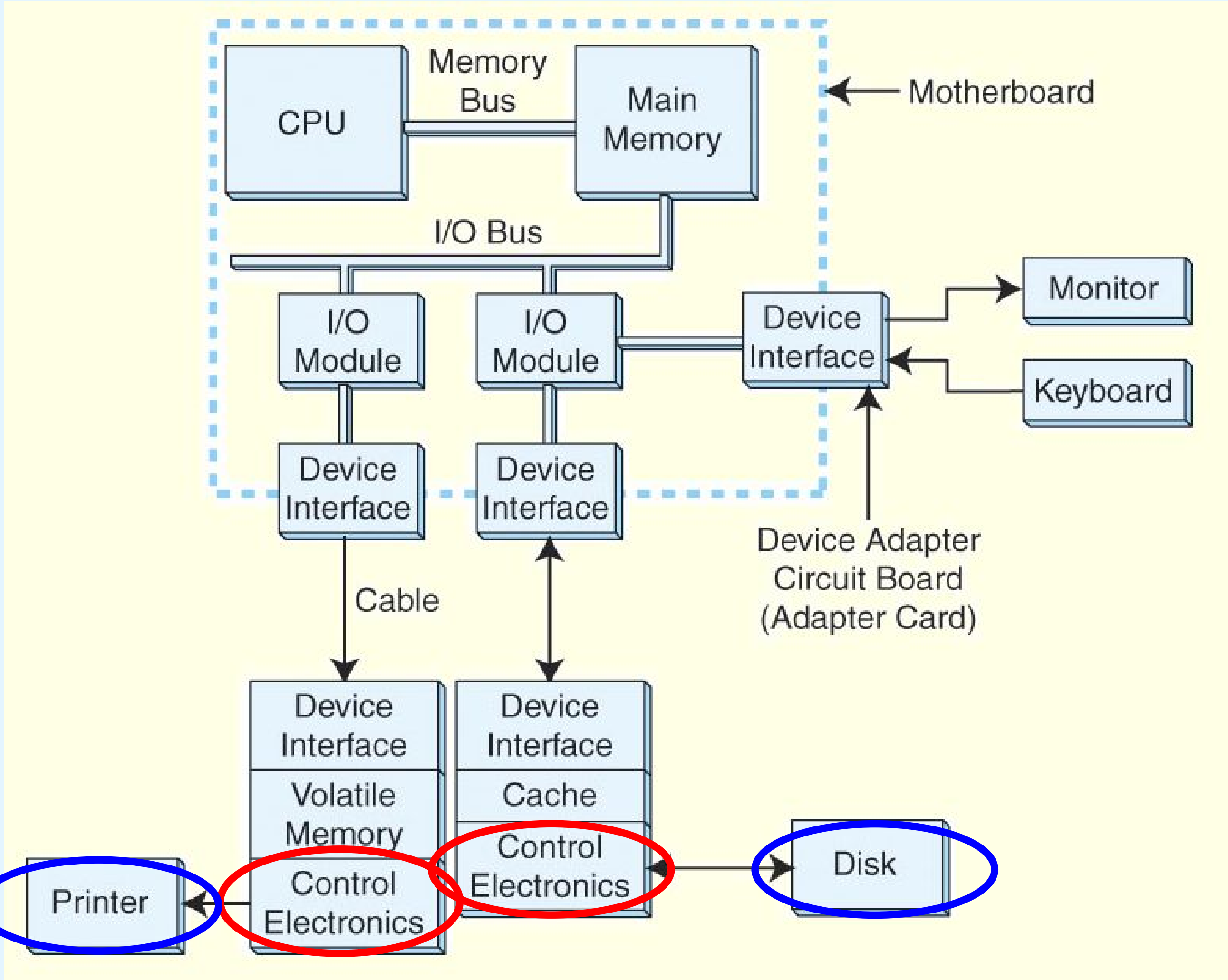
- A model IO configuration (cont.)



– devices that deal with large data blocks have **buffers/caches** that mask slow data transfers from lower level components (e.g., hard drive surfaces)

6.4 I/O Architectures
- I/O Subsystems

- A model IO configuration (cont.)

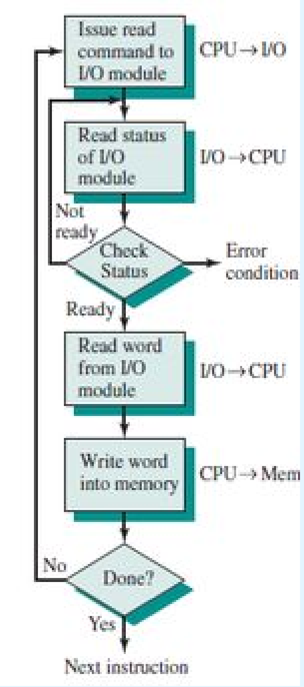


  – **control circuits** in the devices ensure that data is transferred between caches/ buffers and the device

6.4 I/O Architectures
- I/O Control Methods

- I/O between CPU & I/O modules can be controlled in several ways.

1) Programmed I/O (*polled I/O or* ***PIO***): involves periodically querying (polling) for the completion of a previous request



- **port-mapped I/O** (**PMIO**) or **isolated I/O**: I/O modules have a local memory, whose addresses are called **ports**; CPU polls the I/O module and access I/O ports using few specific instructions
- **memory-mapped I/O** (**MMIO**): I/O modules use reserved **main memory blocks**, that are monitored and accessed by the CPU
- poor efficiency: polling is inconsequent most of the times (*)
- simple to implement, suitable to low E/S transfer rates (*)
- allows a *programmatic approach* to I/O control (the programmer chooses when and how many times to poll)

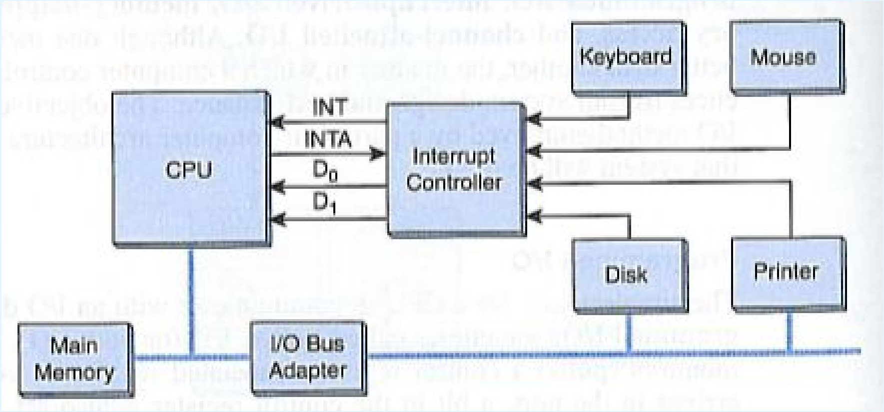
Source: http://inputoutput5822.weebly.com/programmed-io.html

6.4 I/O Architectures
- I/O Control Methods

2) Interrupt Driven I/O: “the reverse of programmed I/O”

- allows the CPU to do other things until I/O is requested
- peripherals interrupt the CPU only when absolutely needed (*)
- flexible: service routines may adapt to hardware changes

I/O subsystem that supports interrupts



(*) When Poll Is Better than Interrupt: https://www.usenix.org/conference/fast12/when-poll-better-interrupt

6.4 I/O Architectures
- I/O Control Methods

3) Direct Memory Access (DMA): offloads I/O processing to a special-purpose chip that takes care of the I/O transfers

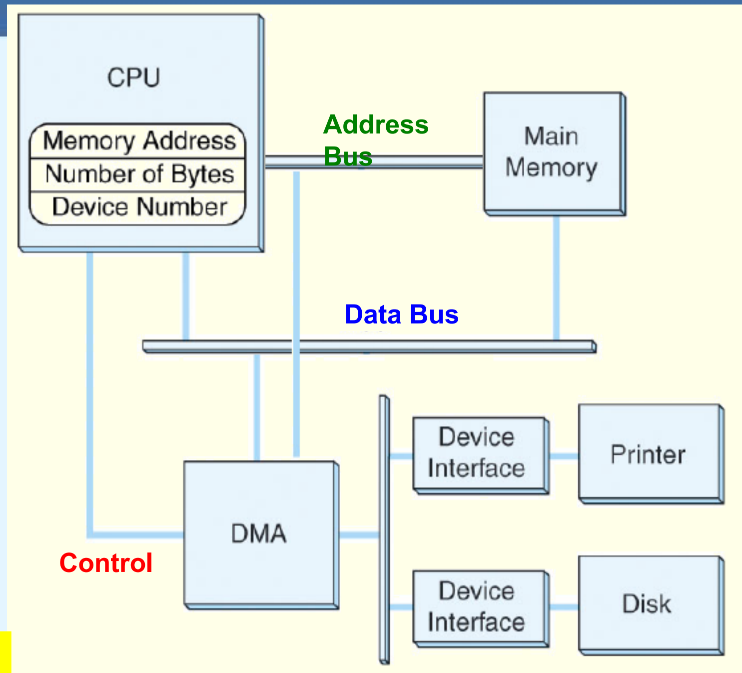
- in the former approaches (pooling and interrupts), it is the CPU that moves data between the devices and main memory:

```
WHILE More-input AND NOT Error
       ADD 1 TO Byte-count
       IF Byte-count > Total-bytes-to-be-transferred THEN
         EXIT
       ENDIF
       Place byte in destination buffer
       Raise byte-ready signal
       Initialize timer
       REPEAT
         WAIT
       UNTIL Byte-acknowledged, Timeout, OR Error
ENDWHILE
```

- this simple algorithm is now executed by a DMA controller

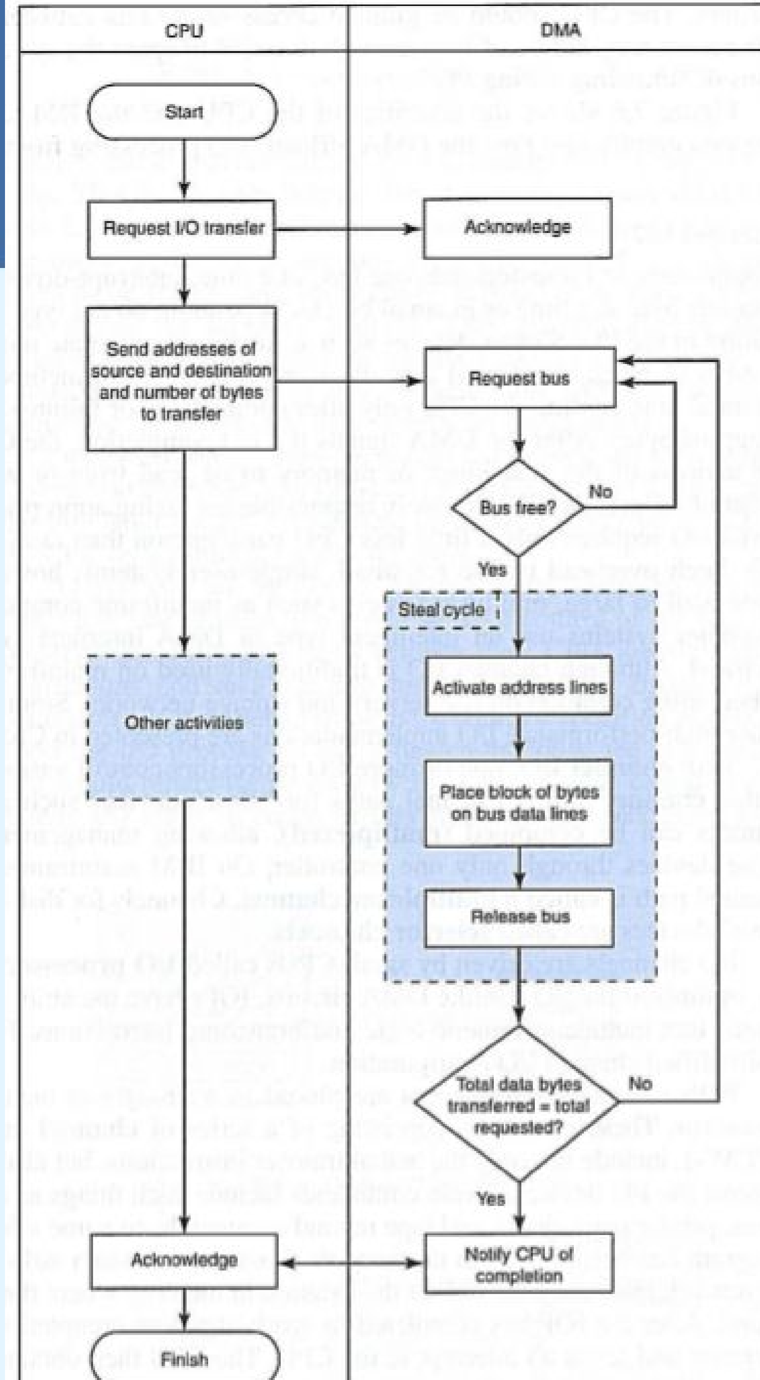
6.4 I/O Architectures
- I/O Control Methods

3) DMA (cont.):



DMA configuration

- CPU tells to the DMA controller the *source*, *destiny* and amount of bytes
- DMA controller works in parallel with CPU, with whom shares the bus, but has higher access priority to avoid I/O timeouts (it “steals” memory cycles from the CPU)



CPU ↔ DMA controller interaction

6.4 I/O Architectures
- I/O Control Methods

4) Channel I/O: uses **dedicated I/O processors** (IOPs)

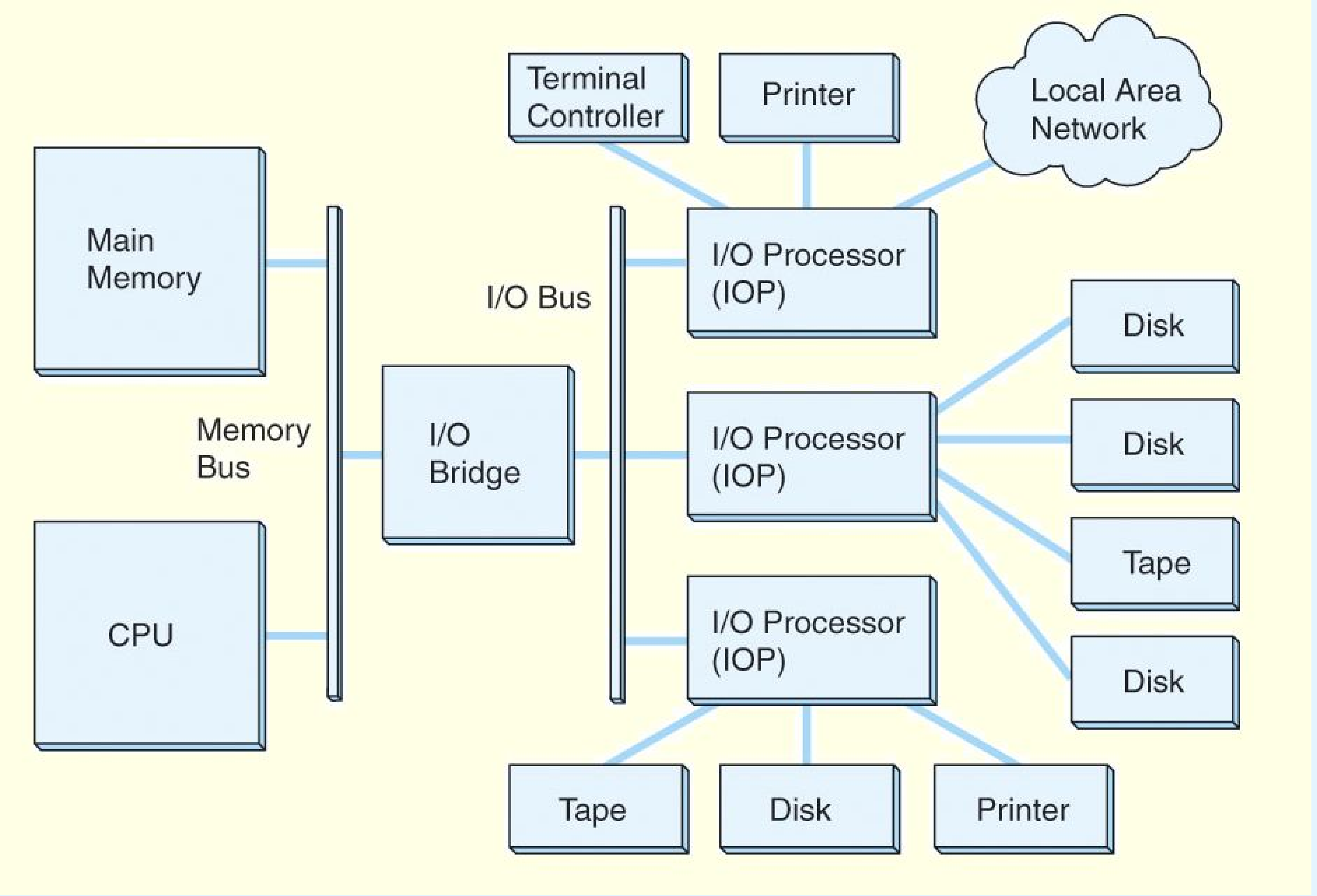
- programmed I/O transfers very small amounts of data; interrupt driven I/O deals with small data blocks; DMA transfers a set of blocks and only interrupts CPU after all blocks are transferred
- in (ancient / mainframe) multi-user systems, performance requisites demand a different approach to I/O control
- channel I/O, based on dedicated I/O processors (IOPs), is different from DMA in that IOPs are more intelligent
- several slow devices (terminals, printers) are multiplexed in a fast channel, managed by an IOP

6.4 I/O Architectures
- I/O Control Methods

checkpoint: exercises ES1, ES2, ES3 and ES4

4) Channel I/O (cont.):

- an IOP negotiates protocols, issues device commands, translates storage coding to memory coding, and can transfer entire files independent of the host CPU (this has only to create the program instructions for the I/O operation and tell the IOP where to find them)

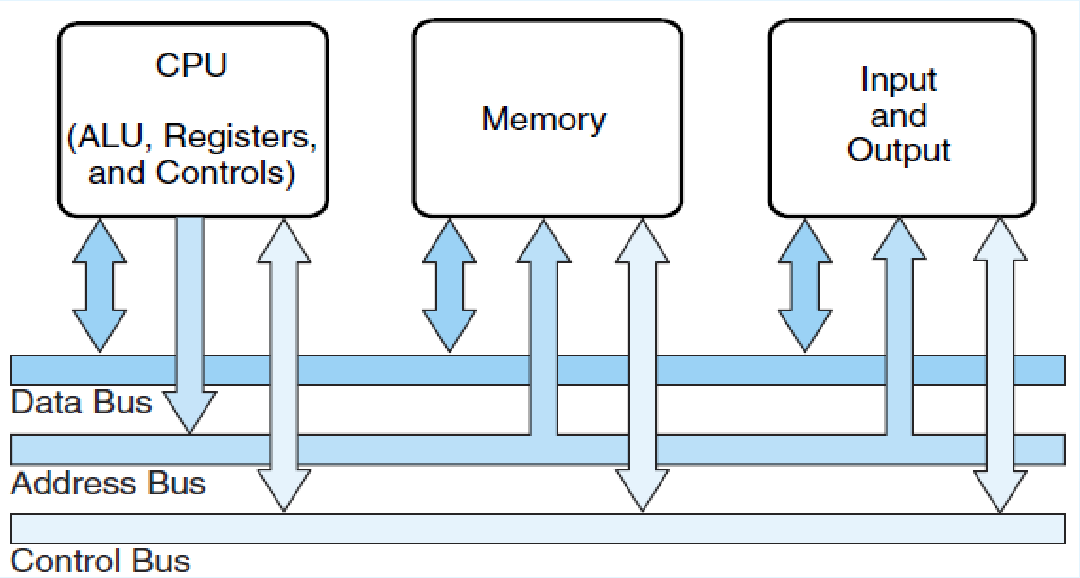


6.4 I/O Architectures
- Character I/O vs Block I/O

- **Character I/O devices**
  - Process/transfer one byte at a time
  - Examples include modems, keyboards, and mice
  - Keyboards are usually connected through an interrupt-driven I/O system. USB mouses use polled I/O
- **Block I/O devices**
  - Process/transfer one block (group of bytes) at a time
  - Most mass storage devices (HDs, Tapes) are block I/O devices
  - Block I/O systems are most efficiently connected through DMA or channel I/O

6.4 I/O Architectures
- I/O Buses Operation

- High level view of the system buses



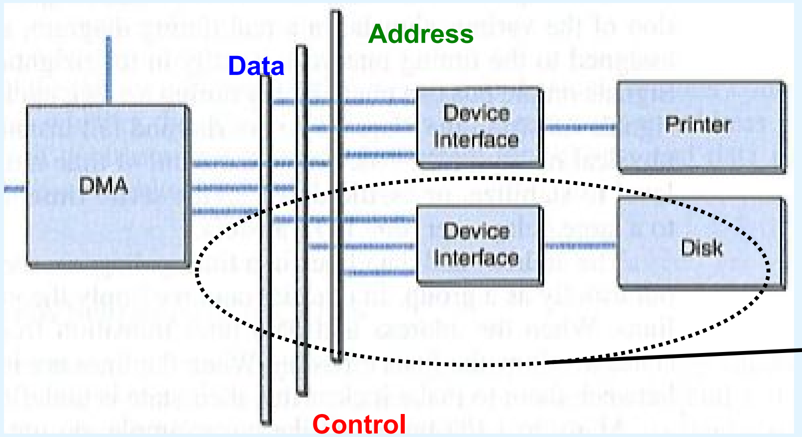
  - bus ***control lines*** activate the devices when they are needed, raise signals when errors have occurred, and reset devices when necessary
  - the **number of data lines** is **the width of the bus**
  - the **address lines** define the source/destiny of data
  - a bus **clock** coordinates activities and provides bit cell boundaries

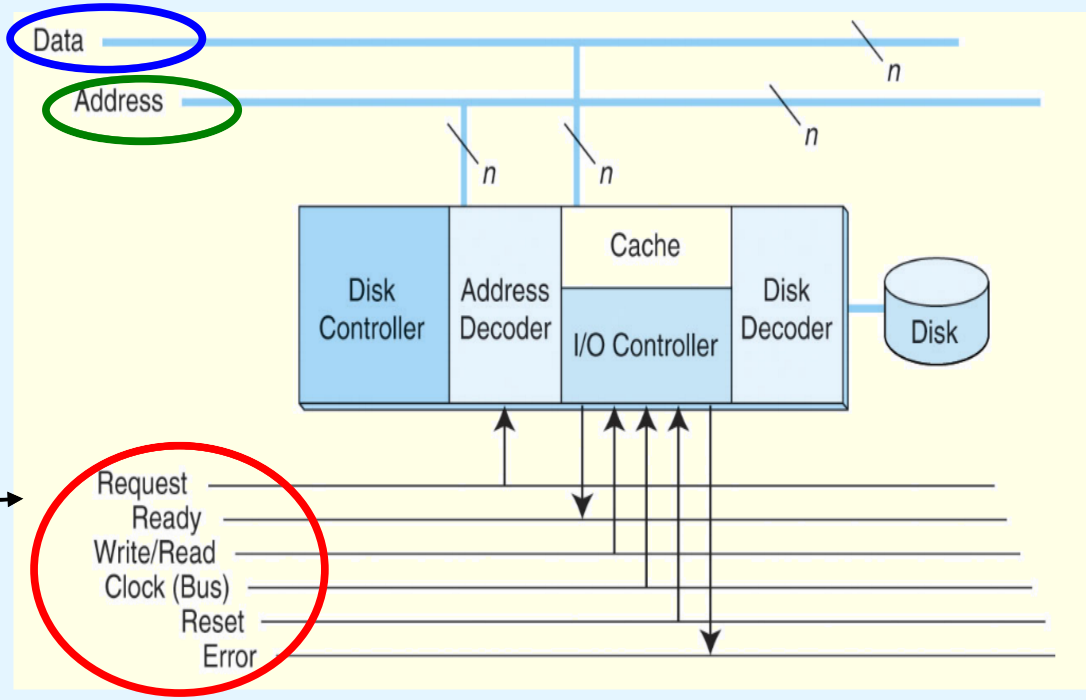
6.4 I/O Architectures
- I/O Buses Operation

- usually, the **memory bus** is separated from the **I/O bus(es)** that link devices to the host system
- the **memory bus** operates **synchronously**
  – sender and receiver share the same clock
- **I/O buses** operate **asynchronously**
  – the protocols used require additional clock(s) for bit timing and management of signals transitions
  – devices connected to the bus may not be ready (*offline*)
  – requests for bus access must be arbitrated among the devices that share the bus

6.4 I/O Architectures
- I/O Buses Operation

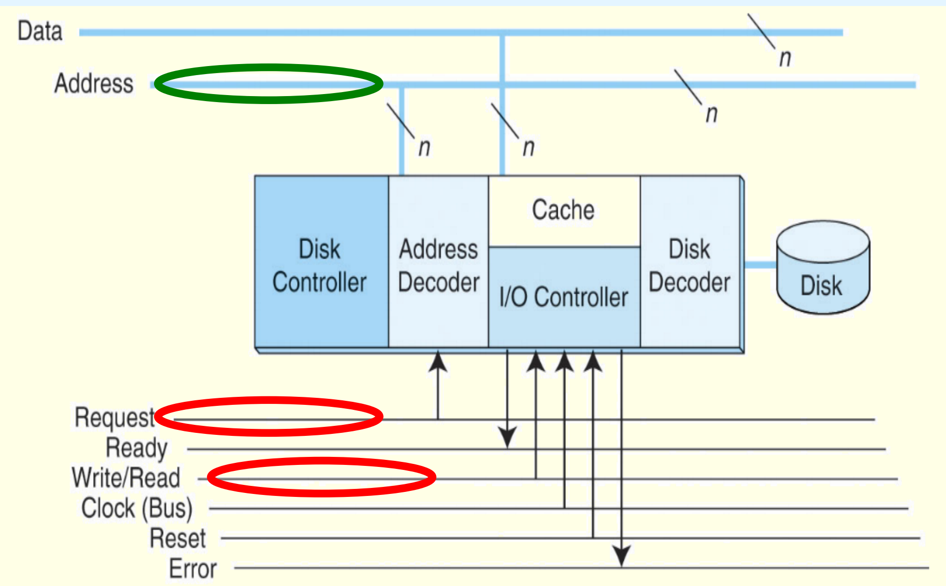
- example: interaction DMA ↔ hard-disk controller
  – three buses: data, addresses and control

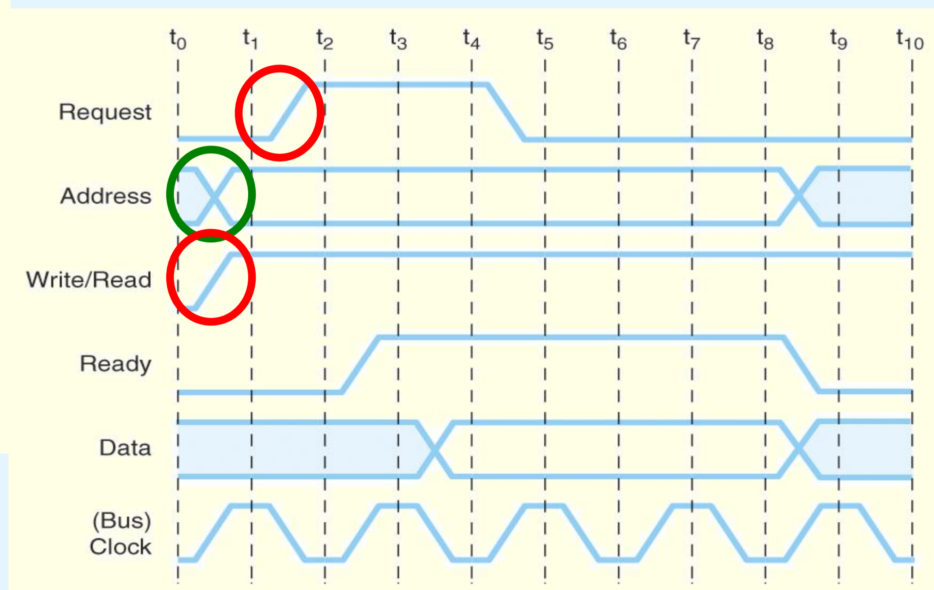




6.4 I/O Architectures
- I/O Buses Operation

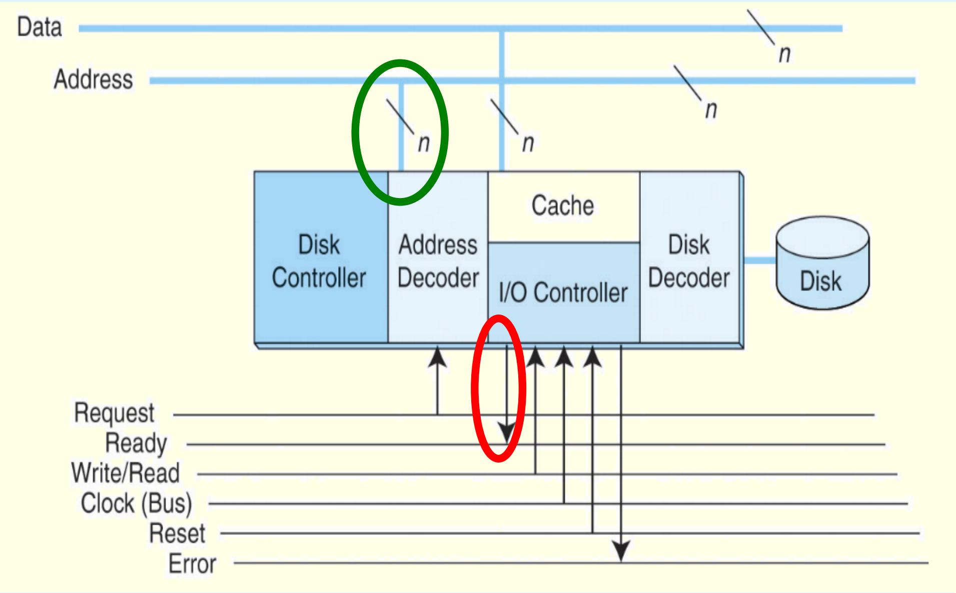
- example: interaction DMA ↔ hard-disk controllers
  - 1) DMA controller: places on the address bus the identifier of the disk controller and activates the Request and Write lines

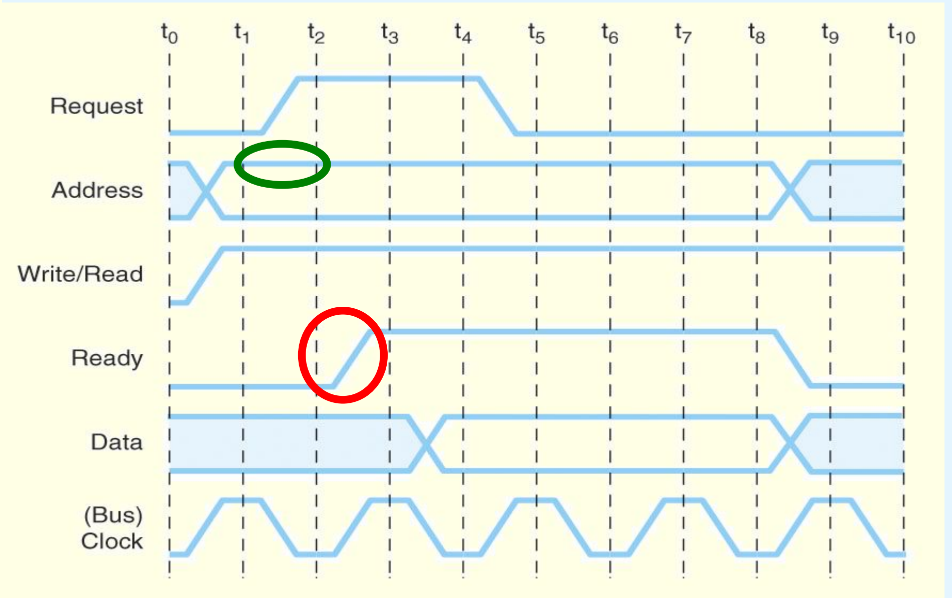




6.4 I/O Architectures
- I/O Buses Operation

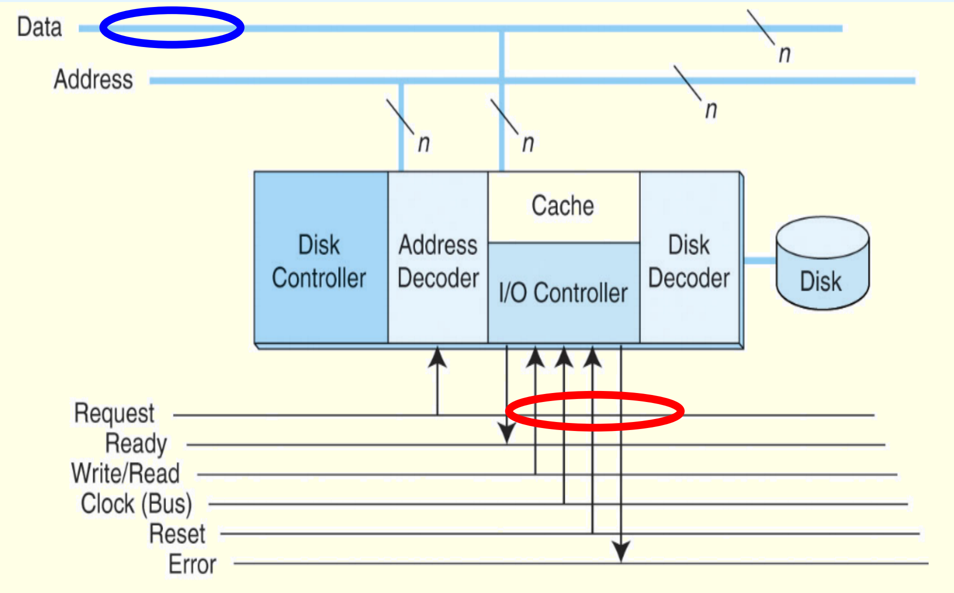
- example: interaction DMA ↔ hard-disk controllers
  - 2) disk controller: as an answer to the Request signal, it checks the address bus; the detected address matches its own address; thus, if the disk is available, it activates the Ready line in the control bus (handshake concluded)

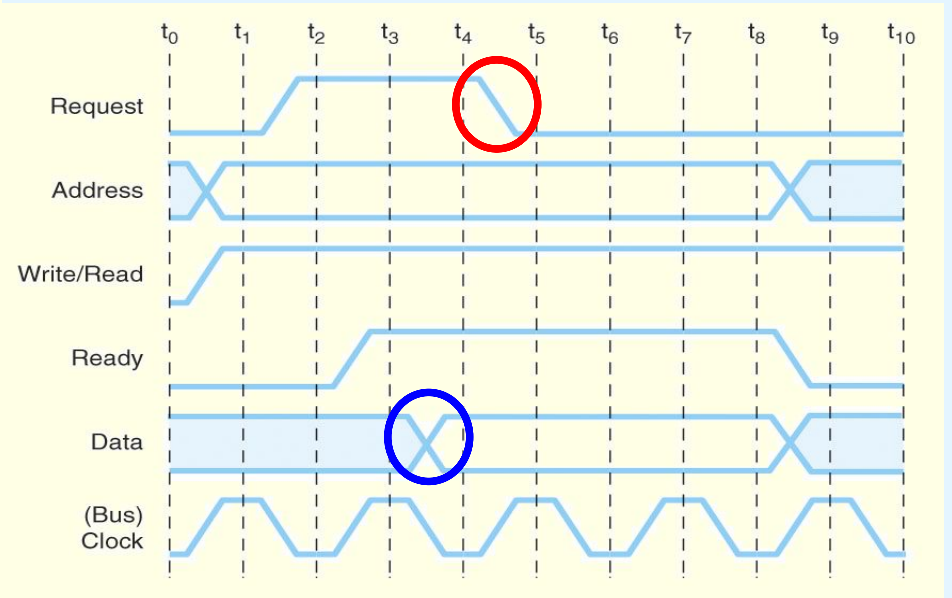




6.4 I/O Architectures
- I/O Buses Operation

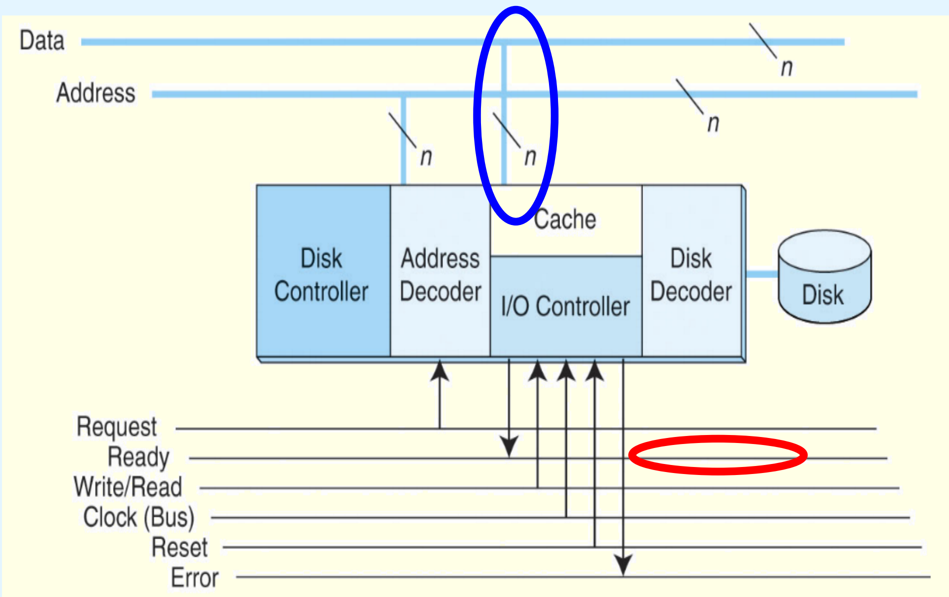
- example: interaction DMA ↔ hard-disk controllers
  - 3) DMA controller: places data in the data bus and deactivates the Request line

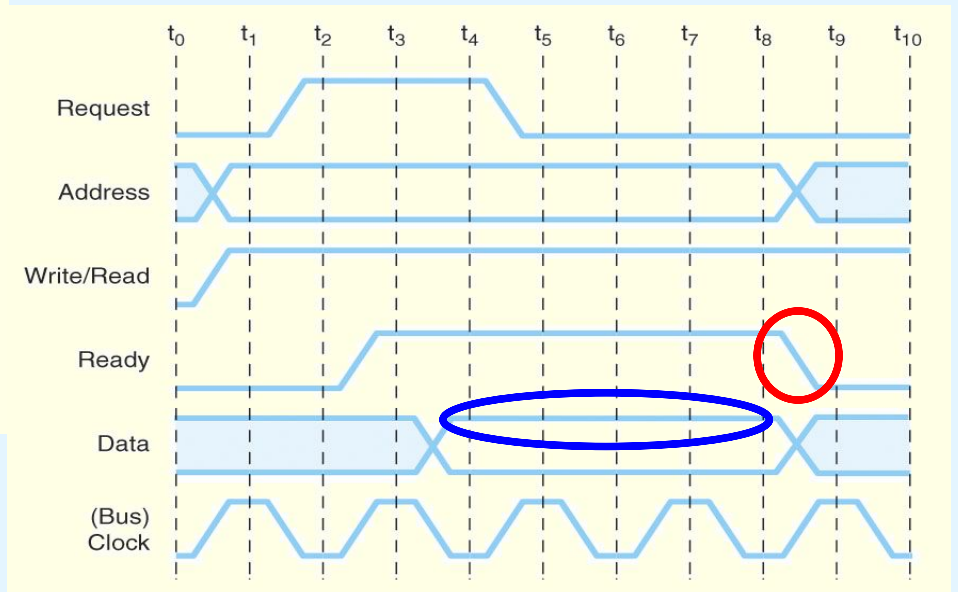




6.4 I/O Architectures
- I/O Buses Operation

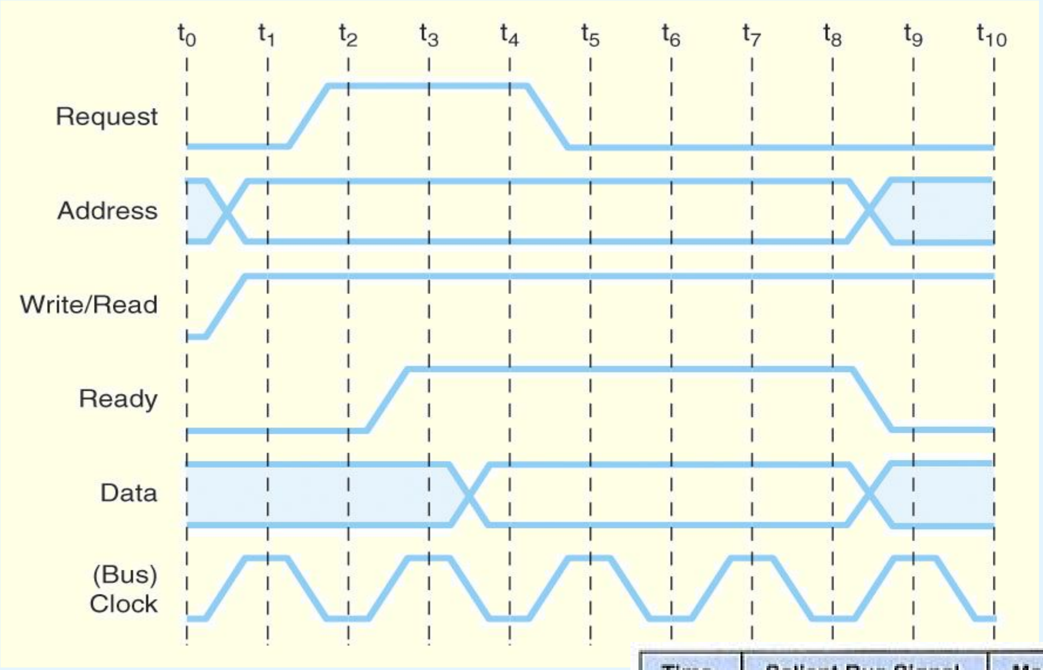
- example: interaction DMA ↔ hard-disk controllers
  - 4) disk controller: transfers data from the data bus to the disk buffer and deactivates the Ready line





6.4 I/O Architectures
- I/O Buses Operation

- example: interaction DMA ↔ hard-disk controllers
  - full timing diagram for the write operation just described



- bus lines can only change in the bus clock transitions

- bus lines don't change instantaneously (there's a settle time)

Time	Salient Bus Signal	Meaning
t_0	Assert Write	Bus is needed for writing (not reading)
t_1	Assert Address	Indicates where bytes will be written
t_2	Assert Request	Request write to address on address lines
t_3	Assert Ready	Acknowledges write request, bytes placed on data lines
t_4–t_7	Data Lines	Write data (requires several cycles)
t_8	Lower Ready	Release bus

6.5 Storage Media

- Secondary Storage

- **secondary storage devices**
  - allow to persistently store large amounts of data
  - slower acess than to RAM, faster acesso than optical medis
  - they are random (or direct) acess devices
    - data stored in **blocks**, all fo the same size
    - each **block** has a **unique address**, and can be acessed **independently** of the others (though usually the neighbor blocks are also read into a private cache located in the device)
      - disk of N blocks seen as an array of blocks 0 to N-1
  - currently, two main technologies coexist
    - **magnetic disks** (also known as hard disks (HDs))
    - **solid state drives** / SSDs (based on FLASH memories)

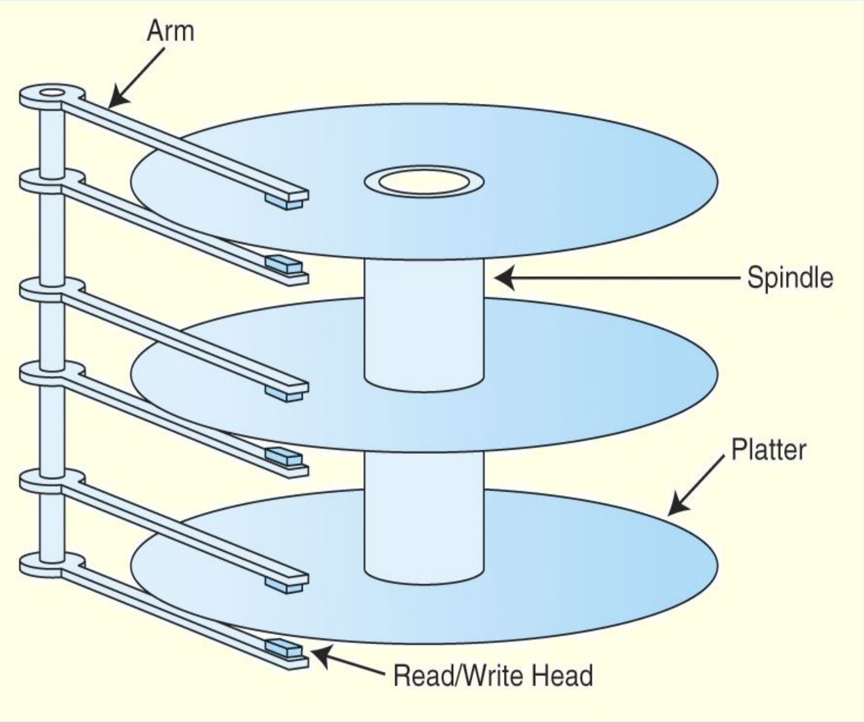
6.5 Storage Media

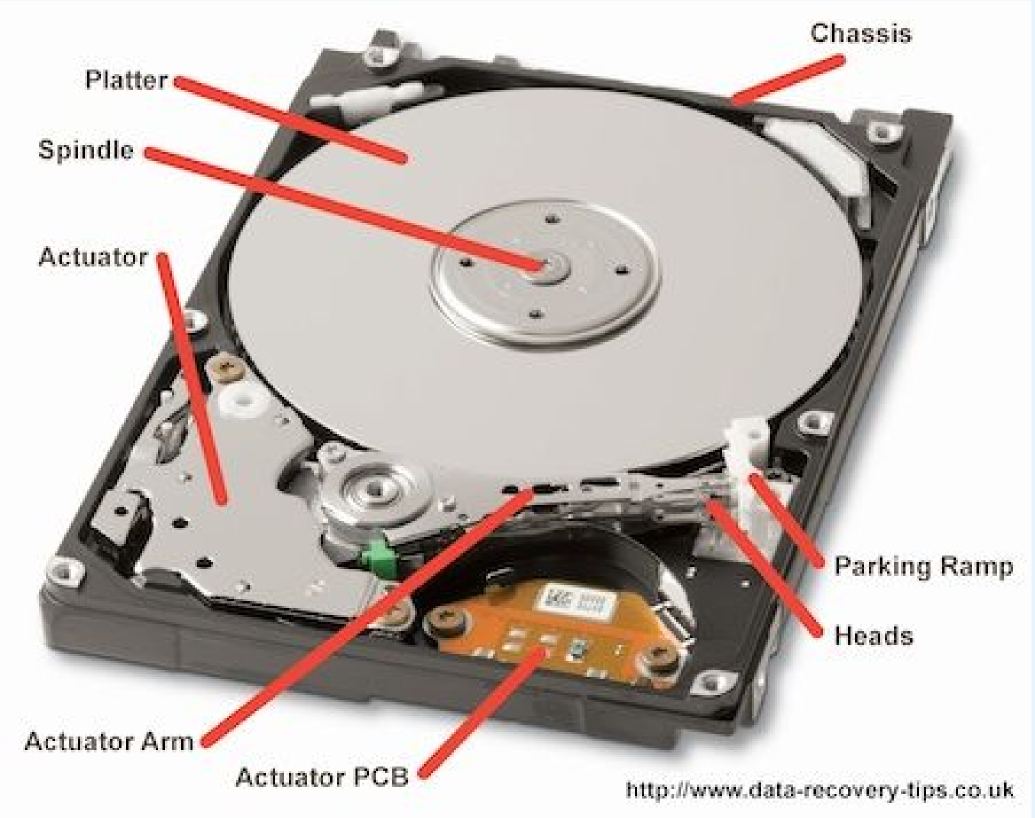
- Secondary Storage: Magnetic Disks

- **Basic Components**
  - **platters** made of non-magnetic material (glass, aluminum alloy), with a metal-particle layer sensitive to magnetization
  - platters mounted on a **spindle**, rotating at
    - 5400rpm, 5900rpm, 7200rpm, 10000rpm, 15000rpm, ...
  - **read-write heads**
    - mounted on a **comb** that swings radially
    - at microscopic distances from the platters
    - these heads work like *magnets*

6.5 Storage Media
- Secondary Storage: Magnetic Disks

- **Basic Components (cont.)**



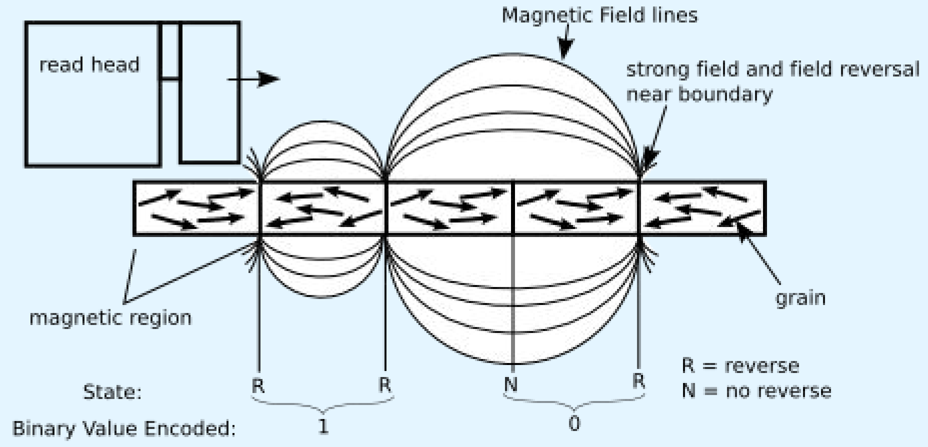


Vídeo: How do Hard Disk Drives Work ?
https://www.youtube.com/watch?v=wtdnatmVdIg

6.5 Storage Media

- Secondary Storage: Magnetic Disks

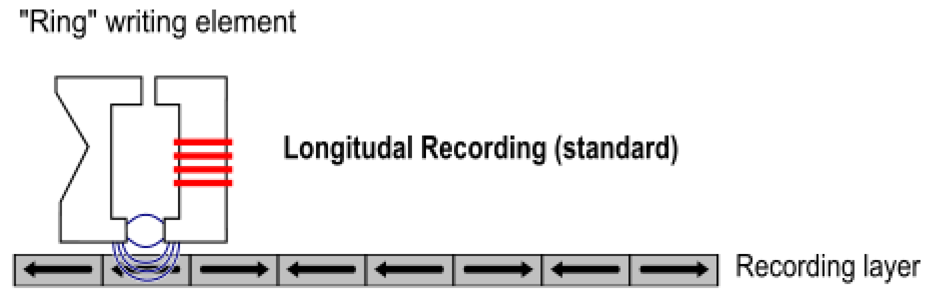
- **Physical Principles**
  - the magnetic surface(s) of each plate include very small size zones (in the order of nanometers) in which the direction of the magnetic field is used to implement a binary code (bits 0/1)
  - the disk heads are able to detect (during read) or change (during writes) the direction of the magnetic field

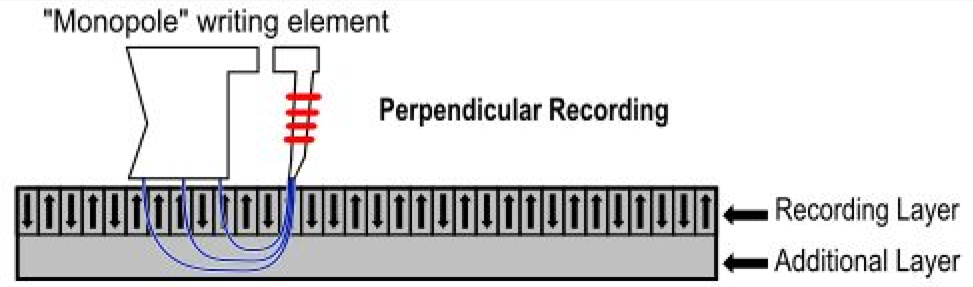


6.5 Storage Media

- Secondary Storage: Magnetic Disks

- **Physical Principles (cont.)**
  - until recently, the magnetic field orientation was horizontal; to improve storage capacity, the orientation became vertical





6.5 Storage Media

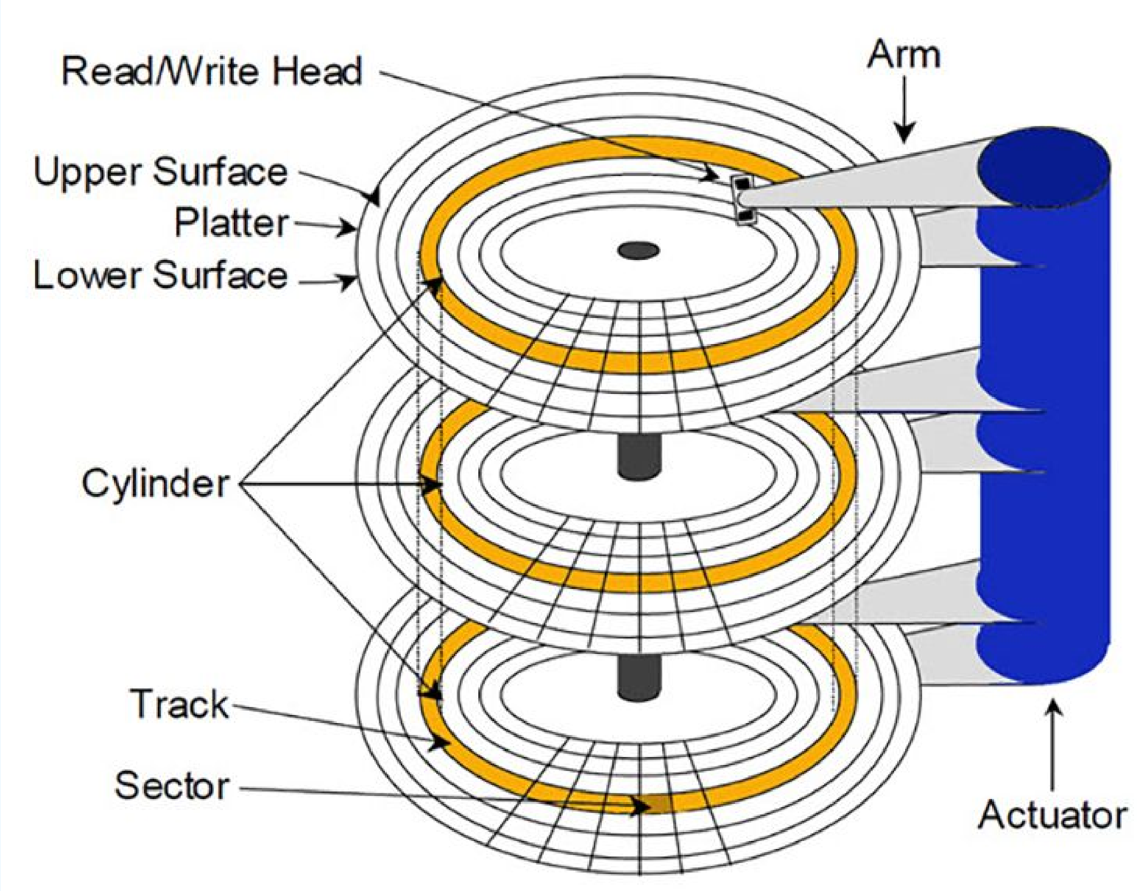
- Secondary Storage: Magnetic Disks

- **Disk Geometry**
  - each surface is divided in **tracks**
  - each track is divided in **sectors** (basic access unit)
    - 512 bytes, 4K bytes (Advanced Format disks)
      - OS access may be to **clusters** (sets of sectors)
    - some storage capacity lost to ECC codes
    - zoned bit recording (**ZBR**): diferent number of sectors per track; tracks group in zones with equal number of sectors
  - the set of tracks in different surfaces, but at the same distance from the center, make a **cylinder**

6.5 Storage Media

- Secondary Storage: Magnetic Disks

- **Disk Geometry**

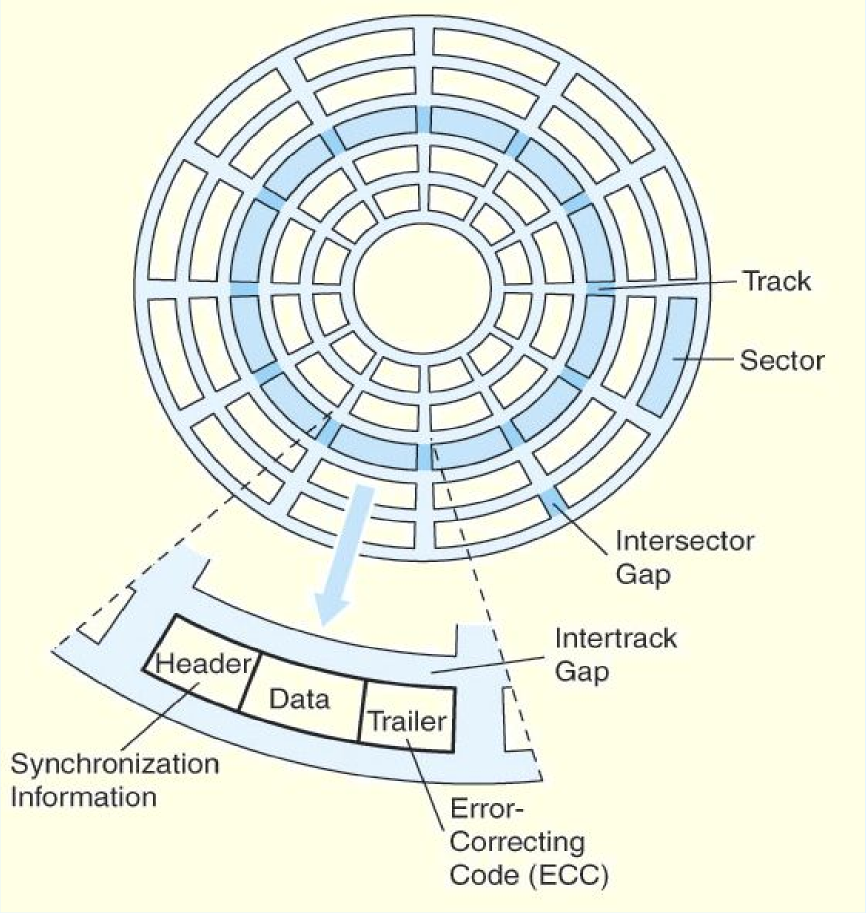


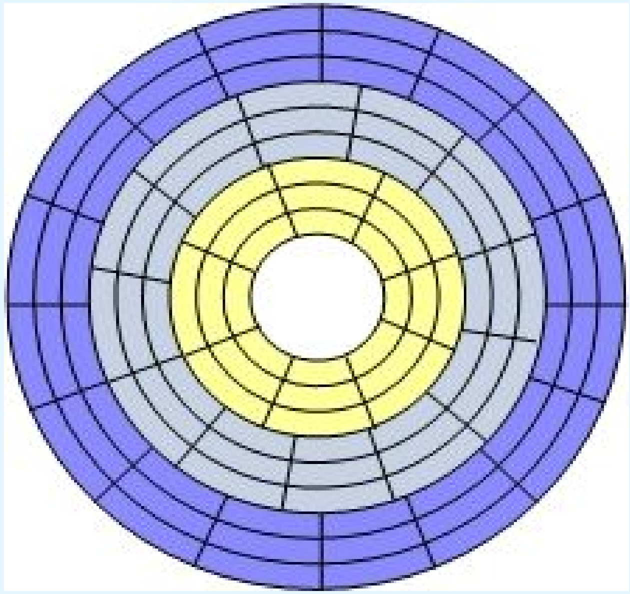
Source: https://www.microcontrollertips.com/can-ssds-replace-storage-memory/

6.5 Storage Media

- Secondary Storage: Magnetic Disks

- **Disk Geometry (cont.)**





zoned bit recording (ZBR);current disks

6.5 Storage Media
- Secondary Storage: Magnetic Disks

- **Disk Geometry**
  - **types of sector coordinates**
    - **linear coordinates: LBA (Logical Block Addressing)**
      - a disk with N sectors is regarded as an array of **N** blocks
      - these are the coordinates used above the controller (SO, apps)
    - **geometric coordinates: CHS** (Cylinder-Head-Sector)
      - reflect the disk geometry; used only by the disk controller
      - in each track, the index of the 1^{st} sector is 1 (not 0 as in LBA)
      - LBA – CHS conversion formulas for non-ZBR disks_

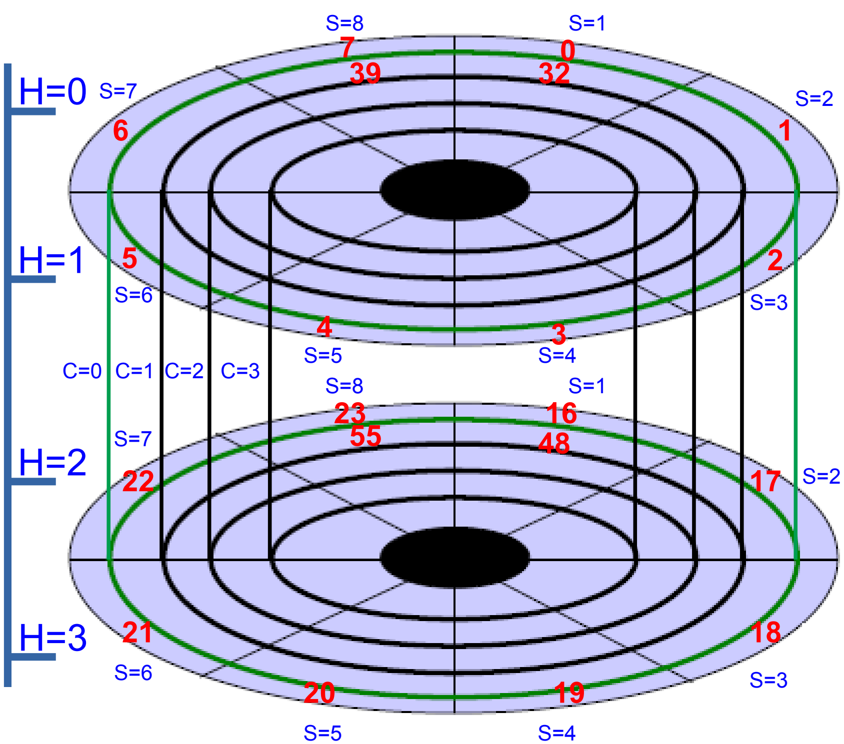
LBA = (C × HPC + H) × SPT + (S − 1)
C = LBA div (HPC × SPT)
H = (LBA div SPT) mod HPC
S = (LBA mod SPT) + 1

LBA = Logical Block Address
C, H, S = cylinder, head, sector number
HPC = Heads Per Cylinder (Surfaces)
SPT = Sectors Per Track

6.5 Storage Media

- Secondary Storage: Magnetic Disks

- **Disk Geometry - example**
  - disk with 2 platters and 2 surfaces per platter; there are 2x2=4 surfaces (and 4 heads, 1 per surface); 4 tracks per surface; **8** sectors per track; thus, there are a total of 4 x 4 x **8** = **128** sectors



distribution of the coordinates LBA 0 to 127:

- starts in the 1st cylinder (**C=0**), 1st head (**H=0**, top surface of the 1st platter), 1st sector (**S=1**): **LBA 0 to LBA 7**

- continue in the 1st cylinder (**C=0**), 2nd head (**H=1**, bottom surface of the 1st platter), 1st sector (**S=1**): **LBA 8 to 15**

- continue in the 1st cylinder (**C=0**), 3rd head (**H=2**, top surface of the 2nd platter), 1st sector (**S=1**): **LBA 16 to 23**

- continue in the 1st cylinder (**C=0**), 4th head (**H=3**, bottom surface of the 2nd platter), 1st sector (**S=1**): **LBA 24 to 31**

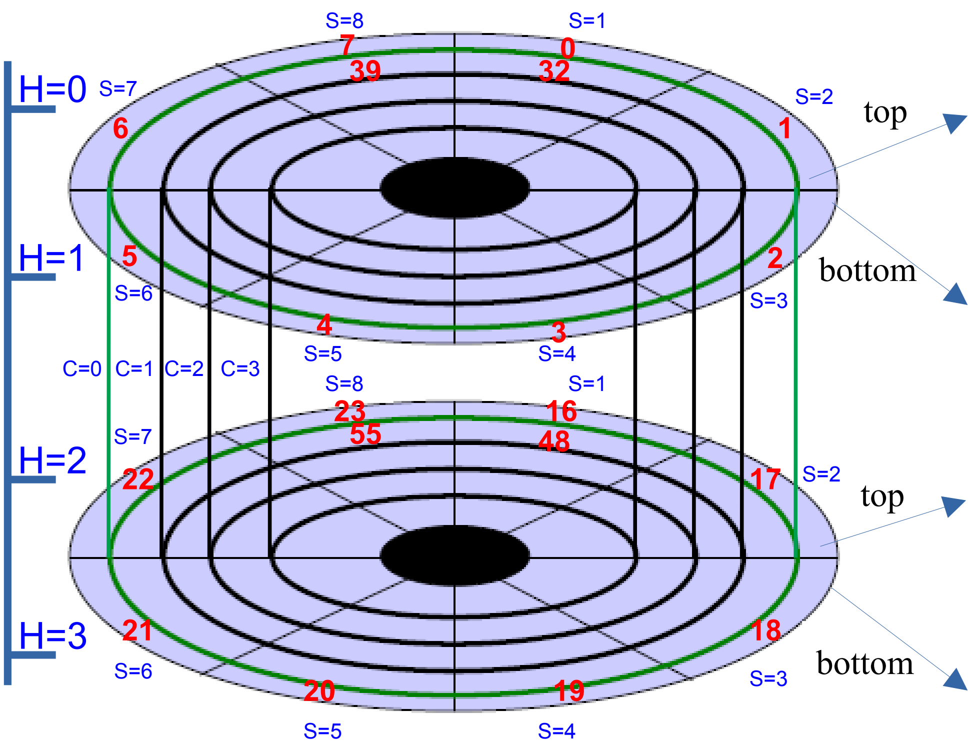
- continue in the 2nd cylinder (**C=1**), 1st head (**H=0**), 1st sector (**S=1**): **LBA 32 to 39**; consume the 2nd cylinder (**LBA 32 to 63**), then continue in the 3rd cylinder (**LBA 64 to 95**) and finally continue to the 4th cylinder (**LBA 96 to 127**)

- the smaller the LBA, the nearest to the 1st cylinder

6.5 Storage Media

- Secondary Storage: Magnetic Disks

- **Disk Geometry - exercise**
  - select some pairs of coordinates (LBA, CHS) and verify the validity of the formulae of slide 37 (note that HPC=4, SPT=8)



LBA – CHS mapping:

cylinder 0				cylinder 1				cylinder 2				cylinder 3			
LBA	C	H	S	LBA	C	H	S	LBA	C	H	S	LBA	C	H	S
0	0	0	1	32	1	0	1	64	2	0	1	96	3	0	1
1	0	0	2	33	1	0	2	65	2	0	2	97	3	0	2
2	0	0	3	34	1	0	3	66	2	0	3	98	3	0	3
3	0	0	4	35	1	0	4	67	2	0	4	99	3	0	4
4	0	0	5	36	1	0	5	68	2	0	5	100	3	0	5
5	0	0	6	37	1	0	6	69	2	0	6	101	3	0	6
6	0	0	7	38	1	0	7	70	2	0	7	102	3	0	7
7	0	0	8	39	1	0	8	71	2	0	8	103	3	0	8
8	0	1	1	40	1	1	1	72	2	1	1	104	3	1	1
9	0	1	2	41	1	1	2	73	2	1	2	105	3	1	2
10	0	1	3	42	1	1	3	74	2	1	3	106	3	1	3
11	0	1	4	43	1	1	4	75	2	1	4	107	3	1	4
12	0	1	5	44	1	1	5	76	2	1	5	108	3	1	5
13	0	1	6	45	1	1	6	77	2	1	6	109	3	1	6
14	0	1	7	46	1	1	7	78	2	1	7	110	3	1	7
15	0	1	8	47	1	1	8	79	2	1	8	111	3	1	8
16	0	2	1	48	1	2	1	80	2	2	1	112	3	2	1
17	0	2	2	49	1	2	2	81	2	2	2	113	3	2	2
18	0	2	3	50	1	2	3	82	2	2	3	114	3	2	3
19	0	2	4	51	1	2	4	83	2	2	4	115	3	2	4
20	0	2	5	52	1	2	5	84	2	2	5	116	3	2	5
21	0	2	6	53	1	2	6	85	2	2	6	117	3	2	6
22	0	2	7	54	1	2	7	86	2	2	7	118	3	2	7
23	0	2	8	55	1	2	8	87	2	2	8	119	3	2	8
24	0	3	1	56	1	3	1	88	2	3	1	120	3	3	1
25	0	3	2	57	1	3	2	89	2	3	2	121	3	3	2
26	0	3	3	58	1	3	3	90	2	3	3	122	3	3	3
27	0	3	4	59	1	3	4	91	2	3	4	123	3	3	4
28	0	3	5	60	1	3	5	92	2	3	5	124	3	3	5
29	0	3	6	61	1	3	6	93	2	3	6	125	3	3	6
30	0	3	7	62	1	3	7	94	2	3	7	126	3	3	7
31	0	3	8	63	1	3	8	95	2	3	8	127	3	3	8

6.5 Storage Media

- Secondary Storage: Magnetic Disks

- **Performance Parameters**
  - There are a number of electromechanical properties of hard disk drives that determine how fast its data can be accessed.
    - **Seek time** is the time that it takes for a disk arm to move into position over the desired cylinder.
    - **Rotational delay** is the time that it takes for the desired sector to move into position beneath the read/write head.
      - the average value depends on the **rotational speed**

$$\frac{\dfrac{60 \text{ seconds}}{\text{disk rotation speed}} \times \dfrac{1000 \text{ ms}}{\text{second}}}{2}$$

example: for a disk with **rotational speed of 7200rpm**, the average rotational delay will be (60 / 7200) / 2 = 4.16ms

6.5 Storage Media
- Secondary Storage: Magnetic Disks

- **Performance Parameters**
  - **Acess Time = Seek time** + (average) **Rotational delay**
  - **Transfer Rate**: rate at which data can be read from the disk
    - max rates: SATA2 3Gb/s 10Krpm ≈ 118 MB/s, SATA3 6Gb/s 10Krpm ≈ 145 MB/s; SAS 6/12 Gb/s 10Krpm ≈ 204 MB/s
- **Other Parameters**
  - **Mean Time To Failure (MTTF):** statistically-determined value calculated experimentally by the manufacturer (intensive use in the lab)
  - **Load/Unload cycles:** number of times that the disk will be turned on/off; should be high for desktop and GREEN disks
  - **Number or Bays:** number of disks that may co-exist in the same chassis; related with the vibration and noise tolerance levels of the disk

6.5 Storage Media

- Secondary Storage: Magnetic Disks

- **Manufacturer Specifications - example**

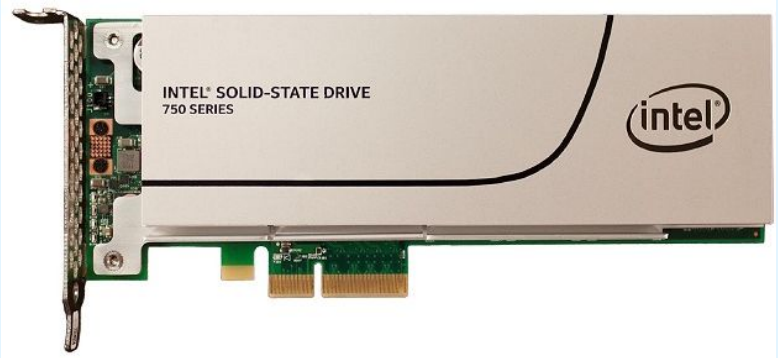
Specifications[1]	750 GB	500 GB	500 GB	320 GB	250 GB	160 GB
Model number	WD7500BPKT	WD5000BPKT	WD5000BEKT	WD3200BEKT	WD2500BEKT	WD1600BEKT
Interface	SATA 3 Gb/s	SATA 3 Gb/s	SATA 3 Gb/s	SATA 3 Gb/s	SATA 3 Gb/s	SATA 3 Gb/s
Formatted capacity	750,156 MB	500,107 MB	500,107 MB	320,072 MB	250,059 MB	160,041 MB
User sectors per drive	1,465,149,168	976,773,168	976,773,168	625,142,448	488,397,168	312,581,808
Form factor	2.5-inch	2.5-inch	2.5-inch	2.5-inch	2.5-inch	2.5-inch
Advanced Format (AF)	Yes	Yes	No	No	No	No
RoHS compliant[2]	Yes	Yes	Yes	Yes	Yes	Yes
Performance						
Data transfer rate (max) Buffer to host Host to/from drive (sustained)	 3 Gb/s 160 MB/s	 3 Gb/s 160 MB/s	 3 Gb/s 154 MB/s	 3 Gb/s 147 MB/s	 3 Gb/s 115 MB/s	 3 Gb/s 115 MB/s
Cache (MB)	16	16	16	16	16	16
Average latency (ms)	4.2	4.2	4.2	4.2	4.2	4.2
Rotational speed (RPM)	7200	7200	7200	7200	7200	7200
Average drive ready time (sec)	4	4	4	4	4	4
Reliability/Data Integrity						
Load/unload cycles[3]	600,000	600,000	600,000	600,000	600,000	600,000
Non-recoverable read errors per bits read	<1 in 10^{14}	<1 in 10^{14}	<1 in 10^{14}	<1 in 10^{14}	<1 in 10^{14}	<1 in 10^{14}
Limited warranty (years)[4]	5	5	5	5	5	5

[1] As used for storage capacity, one megabyte (MB) = one million bytes, one gigabyte (GB) = one billion bytes, and one terabyte (TB) = one trillion bytes. Total accessible capacity varies depending on operating environment. As used for buffer or cache, one megabyte (MB) = 1,048,576 bytes. As used for transfer rate or interface, megabyte per second (MB/s) = one million bytes per second, and gigabit per second (Gb/s) = one billion bits per second. Effective maximum SATA 3 Gb/s transfer rate calculated according to the Serial ATA specification published by the SATA-IO organization as of the date of this specification sheet. Visit www.sata-io.org for details.

checkpoint: exercises
HD1, HD2, HD3 and HD4

6.5 Storage Media
- Solid State Drive (SSD) Technology

- **SSDs Technology**
  - based on non-volatile, FLASH-type memories
  - performance scales with the number of chips used
    - more chips => greater potential parallelism in access
  - various interfaces (SATA, PCIe/NVMe) and form factors (2.5'', M.2, PCIe card, mSATA)



PCIe

2.5'' SATA

2.5'' U.2 (PCIe)

mSATA

M.2 (SATA, PCIe/ NVMe)

6.5 Storage Media
- Solid State Drive (SSD) Technology

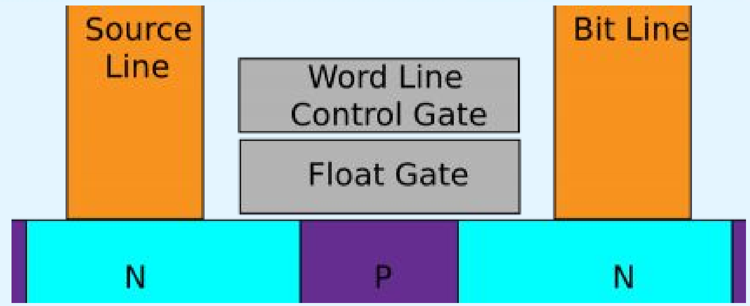
- **SSD Advantages**
  - no moving parts => fast, silent,
    virtually immune to mechanical failure
  - fast startup, constant reading time
  - sustained throughput rates (non RAID)
    - SATA 3Gb/s: read. seq. up-to 280MB/s ; write. seq. up-to 260 MB/s
    - SATA 6Gb/s: read. seq. up-to 550MB/s ; write. seq. up-to 520 MB/s
    - PCIe 3.0 / 4.0 / 5.0:
      - sequential read up-to 3.5 / 5 / 10 GBs
      - sequential write up-to 2.7 / 4.5 / 9,5 GBs
  - performance unaffected by fragmentation

6.5 Storage Media

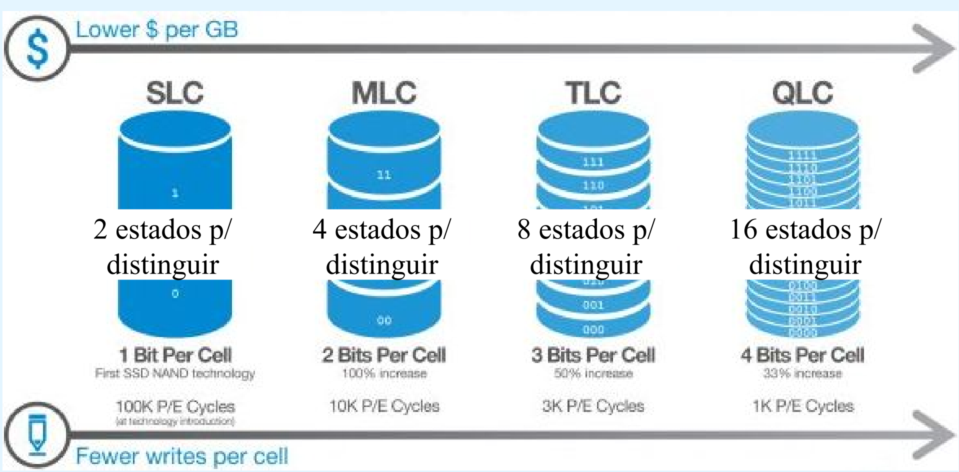
- Solid State Drive (SSD) Technology

- **Disadvantages/Limitations of SSDs**
  - higher cost than magnetic disks
  - lower capacity than magnetic disks
  - support a limited number of (re)writings
    - *wear levelling* => uniform dispersion of writes to the SSD
    - need for suitable File Systems
      (traditional SFs often rewrite the same blocks)
  - R/W asymmetry: writing slower than reading
    - writing performance dependent on free blocks
    - **page**: unity of writing and reading
  - erasing slower than writing
    - **block** (set of pages): erasing unit

6.5 Storage Media
- Solid State Drive (SSD) Technology



- **Basic Components**
  - **flash cell**: the most basic level of storage; electrons on the float gate; on/off charge = 0/1
    - can support 1 (SLC), 2 (MLC), 3 (TLC), or 4 (QLC) bits
      - SLC = Single-Level Cell; MLC = Multi-Level Cell; TLC = Triple-Level Cell; QLC = Quad-Level Cell
    - more bits per cell => lower cost/performance/endurance



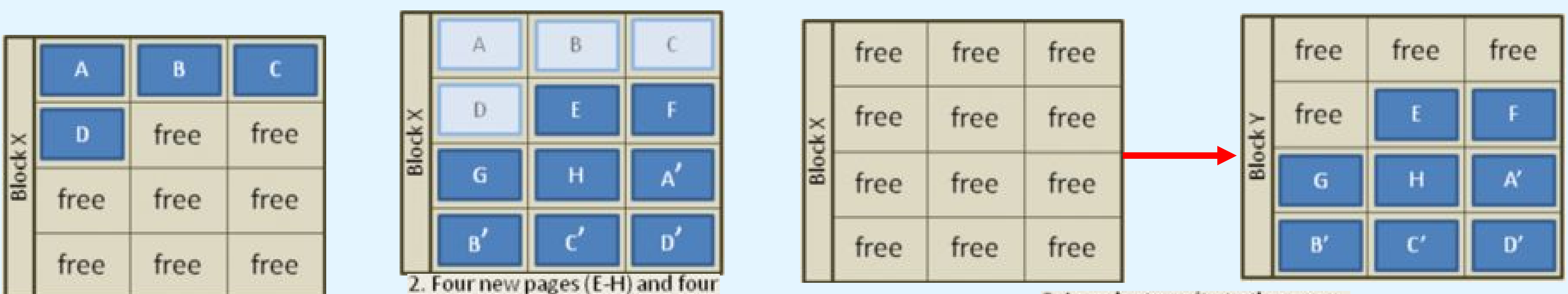
P/E=Program/Erase	SLC	MLC	TLC	HDD	RAM
P/E cycles	100k	10k	5k	*	*
Bits per cell	1	2	3	*	*
Seek latency (μs)	*	*	*	9000	*
Read latency (μs)	25	50	100	2000-7000	0.04-0.1
Write latency (μs)	250	900	1500	2000-7000	0.04-0.1
Erase latency (μs)	1500	3000	5000	*	*

6.5 Storage Media
- Solid State Drive (SSD) Technology

- **Storage Organization**
  - cannot access (**R**/**W**/**D**) an individual **flash cell**
  - **flash cells** are organized in **pages**(2/4/8/16 KB)
  - **pages** are organized in **blocks** (256 KB a 4 MB)
- **Read**: reading unit is the **page;** reading is **very fast**
- **Write/Modify**: writing unit is the **page**
  - **writing** on an **empty page** is **slower than reading the page**
  - **modifying** a **page** is **slower than writing the first time**
    - in practice, a page cannot be modified directly (in place)
    - to modify the content of a page it is necessary to read it into a cache where it will be modified, **invalidate** the original page, and write the modified content to a free page (this page can be in the same block as the original, or in another one)

6.5 Storage Media
- Solid State Drive (SSD) Technology

- **Write/Modify** (cont.): if there are no free pages (or invalid that can become free), we must erase and rewrite the block
- **Delete/Remove**: the deleting **unit** is the **block**
  - to make an **invalid** page free we must delete it
  - we cannot delete individual pages, only blocks
  - **garbage collection**: process triggered by the SSD controller in order to recover free blocks from the invalid ones



1. Four pages (A-D) are written to a block (X). Individual pages can be written at any time if they are currently free (erased).

2. Four new pages (E-H) and four replacement pages (A′-D′) are written to the block (X). The original A-D pages are now invalid (stale) data, but cannot be overwritten until the whole block is erased.

3. In order to write to the pages with stale data (A-D) all good pages (E-H & A′-D′) are read and written to a new block (Y) then the old block (X) is erased. This last step is *garbage collection*.

6.5 Storage Media
- Solid State Drive (SSD) Technology

checkpoint:
exercise SSD1

- **Delete/Remove** (cont.):
  - **TRIM command**:
    - when a file is deleted on a hard disk, the blocks that the file occupies on the disk are not overwritten with zeros
    - these blocks are only marked free by the OS, in the filesystem data structures (which makes *undelete* easier)
    - later, these blocks can be rewritten; the disk doesn't know (and doesn't need to) whether the blocks are free or occupied
    - but an SSD needs to know if certain pages are considered free by the OS, to do garbage collection
      - this notification is done with the TRIM command
      - allows to retrieve free blocks in advance (and not only when they are needed, which would be painful)

6.5 Storage Media

- Solid State Drive (SSD) Technology

- **Videos**:
  - **SSD vs Hard Drive vs Hybrid Drive**
    - https://www.youtube.com/watch?v=1cyMTI_QXSc
  - **M.2 NVMe SSD Explained - M.2 vs SSD**
    - https://www.youtube.com/watch?v=HvfleTieXOI
  - **Inside the Engineering of Solid-State Drive Architecture**
    - https://www.youtube.com/watch?v=r-SivgEpA1Q
  - How do SSDs Work:
    - https://www.youtube.com/watch?v=5Mh3o886qpg
  - The Engineering Puzzle of Storing Trillions of Bits in your Smartphone / SSD
    - https://www.youtube.com/watch?v=5f2xOxRGKqk
  - How does NAND Flash Work? Reading from TLC : Triple Level Cells
    - https://www.youtube.com/watch?v=YtBysgPOKx4

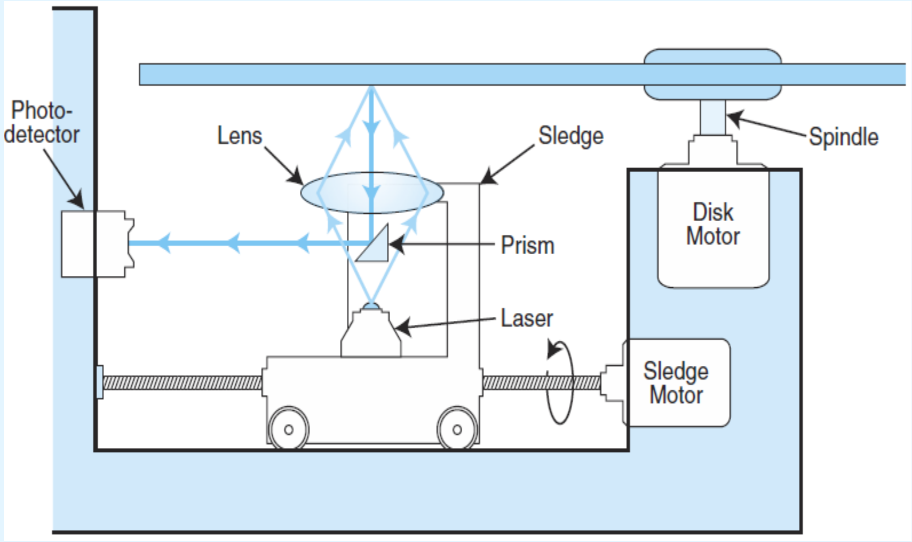
6.5 Storage Media

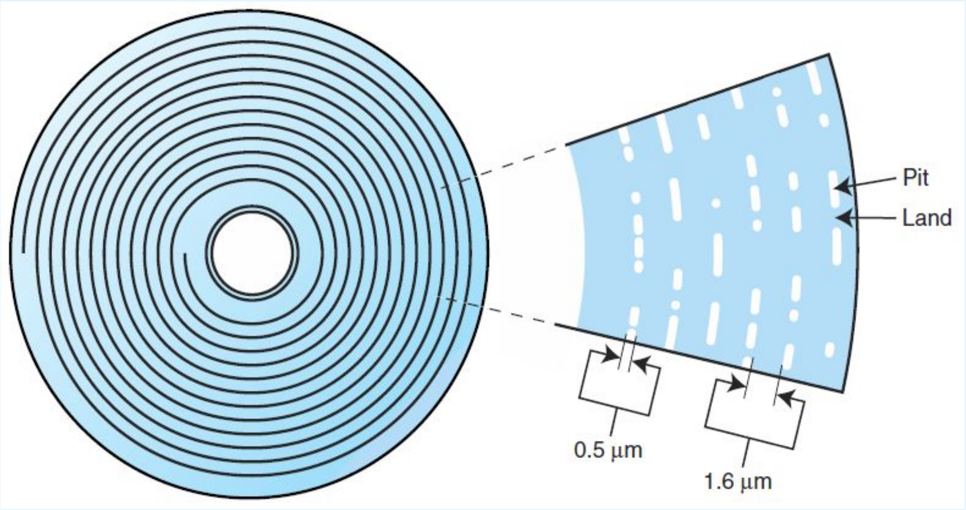
- Optical Discs

- provide large storage capacities very inexpensively
- it is estimated they can endure for a hundred years
- they come in a number of varieties including:
  – CD [-ROM, -R, -RW] ; 650MB, 700MB
  – DVD [-ROM, -R, +R, -RAM] ; 4.37GB, 15.90GB
  – HD-DVD [-ROM, -R, -RW]; 15GB, 30GB
  – BD [-ROM, -R, -RW] ; 25GB, 50GB
- **Optical Media (Video):**
  – https://www.youtube.com/watch?v=UuSLGnqCWdk
- **Optical Drives (Video):**
  – https://www.youtube.com/watch?v=4nORstVKWYI

6.5 Storage Media
- Optical Discs

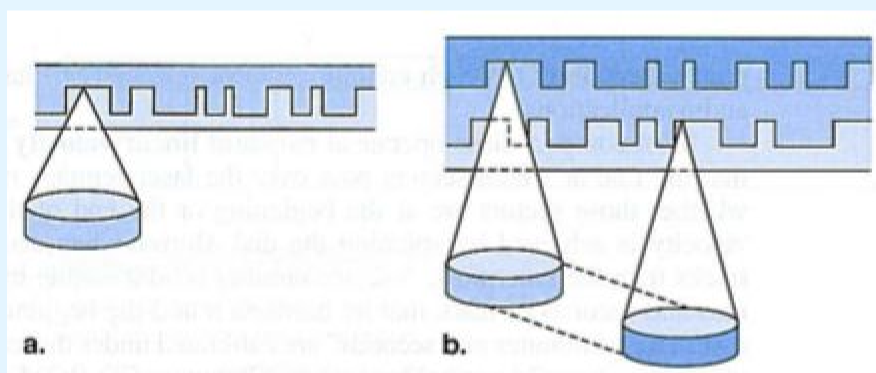
- **CD-ROMs**
  - designed to store music in the 1980s; later adapted to data
  - data is recorded in a single spiral track, from the center outward
  - 1's and 0's are delineated by bumps in the polycarbonate disk substrate; the transitions between pits and lands define binary 1's
  - when reading, the laser reflects differently on the prism for 0's and 1's; the photo-detector makes the conversion to voltage levels





6.5 Storage Media
- Optical Discs

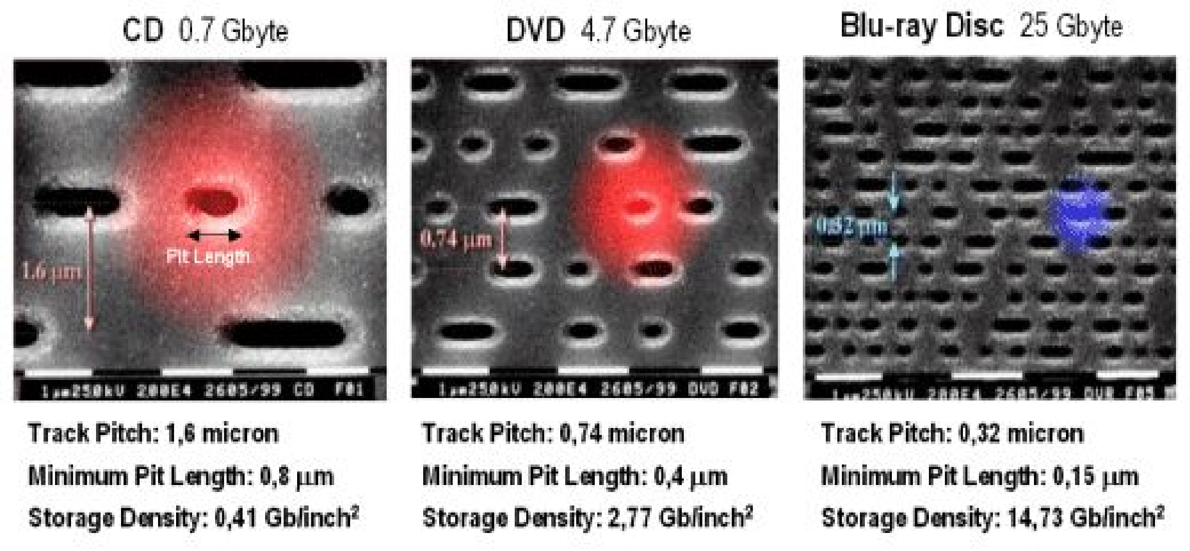
- **DVDs** (Digital Versatile Disk / Digital Video Disk)
  - may store video, audio, or data
  - varieties include **single sided single layer (4.37GB)**, single sided double layer, double sided single layer, and **double sided double layer (15.9 GB)**
  - DVDs employ a laser that has a shorter wavelength than the CD’s laser; this allows pits and land to be closer together and the spiral track to be wound tighter.



a) single-layer vs b) double-layer

6.5 Storage Media
- Optical Discs

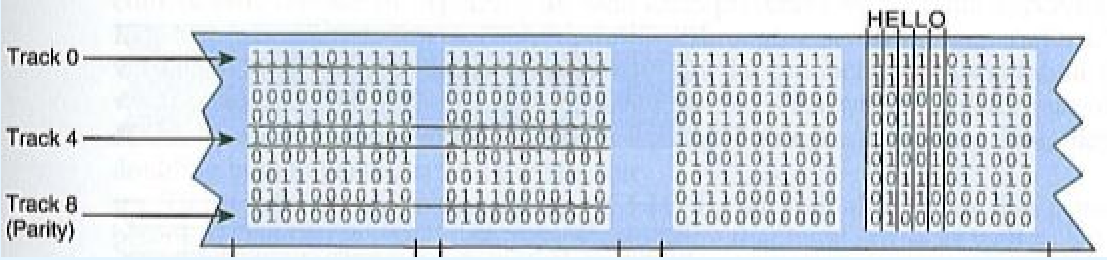
- **Blue-Violet Laser Disks**
  - employ a shorter wavelength than DVDs
  - conceived to store High-Definition content, and data
  - two initial incompatible formats, HD-DVD and Blu-Ray
  - HD-DVD (15GB) supported DVD (red laser) compatibility
  - Blu-Ray offered more storage capacity (single-layer supports 25GB; double layer reaches 50GB); it won the format war



6.5 Storage Media
- Magnetic Tape

- the oldest mass storage support still used, and the cheapest
- first-generation magnetic tape was not much more than wide **analog** recording tape, having capacities under 11MB.

- data was usually written in nine vertical tracks:



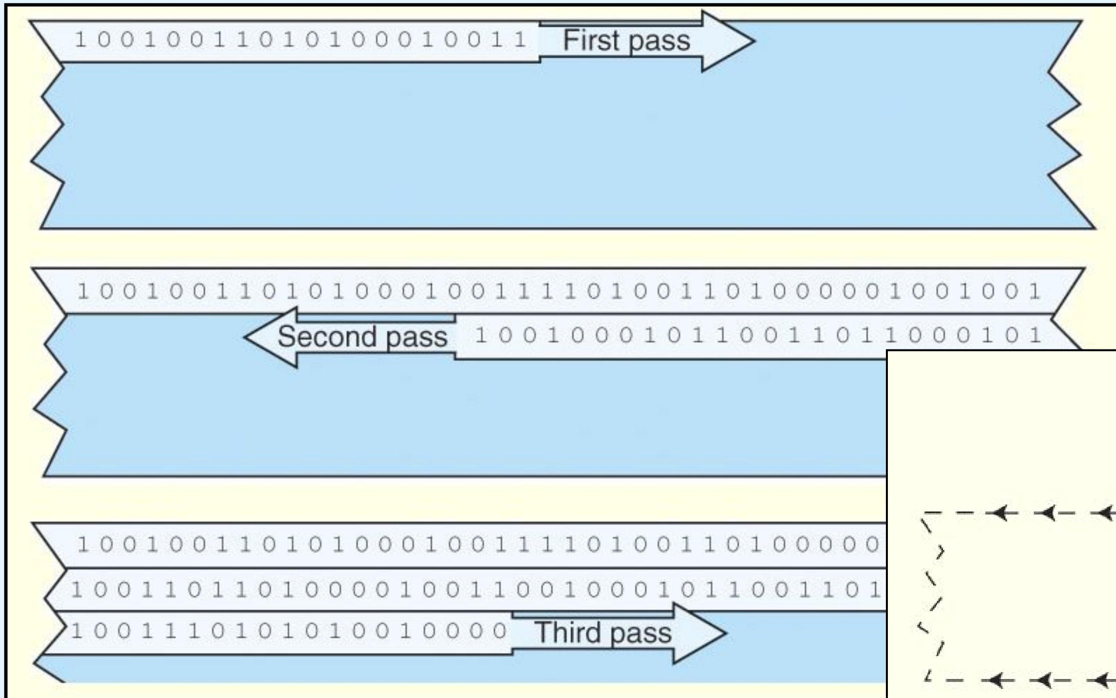
6.5 Storage Media
- Magnetic Tape

- Today’s tapes are **digital**, and provide multiple gigabytes of data storage.
- Two dominant recording methods are ***serpentine*** and ***helical scan***, which are distinguished by how the read-write head passes over the recording medium.
  - Serpentine recording is used in *digital linear tape* (DLT) and *Quarter inch cartridge* (QIC) tape systems.
  - *Digital audio tape* (DAT) systems employ helical scan recording (slower and more abrasive than serpentine)

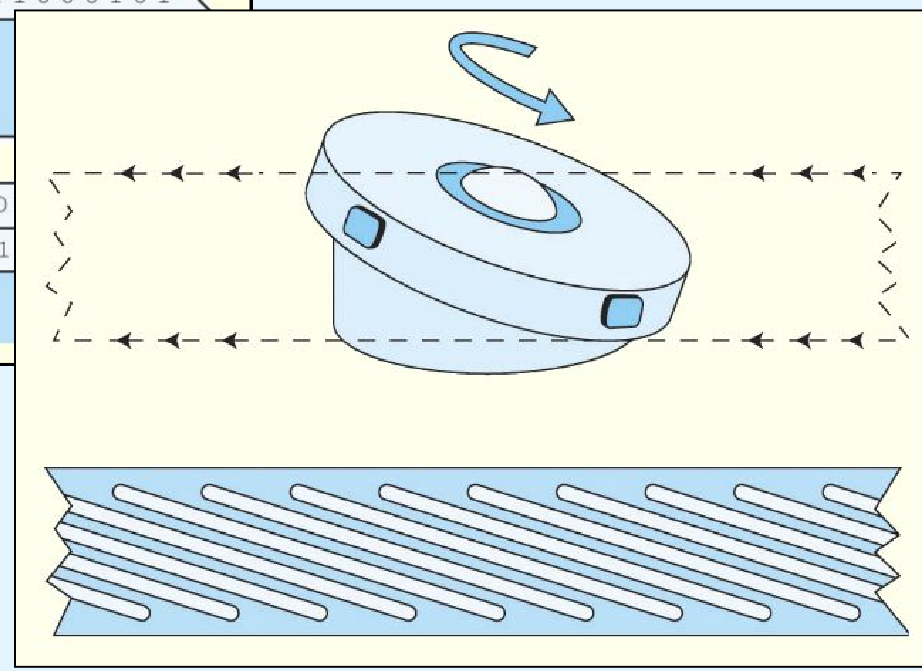
These two recording methods are shown on the next slide.

6.5 Storage Media

- Magnetic Tape



Serpentine



Helical Scan

6.5 Storage Media
- Magnetic Tape

- Numerous incompatible tape formats emerged over the years.
  - Sometimes even different models of the same manufacturer's tape drives were incompatible!
- Finally, in 1997, HP, IBM, and Seagate collaboratively invented a best-of-breed digital tape standard.
- They called this new tape format *Linear Tape Open* (*LTO*) because the specification is openly available.

6.5 Storage Media
- Magnetic Tape

- LTO supports several levels of error correction, providing superb reliability (tape has a reputation for being an error-prone medium)
- The specification allowed for the refinement of the technology through several "generations."

Format	LTO-1	LTO-2	LTO-3	LTO-4	LTO-5	LTO-6	LTO-7	*Type M (M8)*[Note 1]	LTO-8	LTO-9	LTO-10	LTO-11	LTO-12	LTO-13	LTO-14
Release date	2000[12]	2003	2005	2007	2010[13]	Dec. 2012[14]	Dec. 2015[15][16][17]	Dec. 2017		Sep. 2021[18]	*TBA*	*TBA*	*TBA*	*TBA*	*TBA*
Uncompressed/Native capacity	100 GB	200 GB	400 GB	800 GB	1.5 TB[19]	2.5 TB[20]	6.0 TB[17][21]	9 TB	12 TB[22]	18 TB[23][19][24]	36 TB[25]	72 TB[25]	144 TB[25]	288 TB[11]	576 TB[11]
Compressed capacity	200 GB	400 GB	800 GB	1.6 TB	3.0 TB	6.25 TB	15 TB	22.5 TB	30 TB	45 TB	90 TB	180 TB	360 TB	720 TB	1,440 TB
Max uncompressed speed (MB/s)[21][Note 2]	20	40	80	120	140	160	300[26]		360	400	*1,100*	*TBA*	*TBA*	*TBA*	*TBA*
Max compressed speed (MB/s)	40	80	160	240	280	400	750		900	1,000	*2,750*	*TBA*	*TBA*	*TBA*	*TBA*
Time to write a full tape at max speed(hh:mm)	1:23			1:51	3:10	4:20	5:33	8:20	9:16	12:30	*12:07*	*TBA*	*TBA*	*TBA*	*TBA*
Compression capable?	Yes, "2:1"					Yes, "2.5:1"					Planned, "2.5:1"[24][27]				
WORM capable?	No		Yes												
Encryption capable?	No			Yes											
LTFS capable?	No				Yes										
Max. number of partitions	1 (no partitioning)				2										

Source: http://en.wikipedia.org/wiki/Linear_Tape-Open

LTO-9 Introduction (Video): https://www.youtube.com/watch?v=iQXdwr7dJOk
Tape is here to rescue big data (Video): https://www.youtube.com/watch?v=z4DP0Lg-qpw

Conclusion

- Buses require control lines, a clock, and data lines. Timing diagrams specify operational details.
- I/O systems are critical to the overall performance of a computer system.
- I/O systems consist of memory blocks, cabling, control circuitry, interfaces, and media.
- I/O control methods include programmed I/O, interrupt-based I/O, DMA, and channel I/O.

Conclusion

- Magnetic disk is the principal form of durable storage.
- Disk performance metrics include seek time, rotational delay, and reliability estimates.
- Optical disks provide long-term storage for large amounts of data, although access is slow.
- Magnetic tape is also an archival medium. Recording methods are track-based, serpentine, and helical scan.